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Microelectronic devices, and related memory devices and electronic systems

Abstract

A microelectronic device comprises a stack structure, pillar structures, a conductive plug structure, a sense transistor, and selector transistors. The stack structure comprises a vertically alternating sequence of conductive material and insulative material, and is divided into blocks separated by dielectric slot structures. The blocks individually include sub-blocks horizontally extending in parallel with one another. The pillar structures vertically extend through one of the blocks of the stack structure. Each pillar structure of a group of the pillar structures is positioned within a different one of the sub-blocks of the one of the blocks than each other pillar structure of the group. The conductive plug structure is coupled to multiple of the pillar structures of the group of the pillar structures. The sense transistor is gated by the conductive plug structure. The selector transistors couple the sense transistor to a read source line structure and a digit line structure.

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Background/Summary

TECHNICAL FIELD

(1) The disclosure, in various embodiments, relates generally to the field of microelectronic device design and fabrication. More specifically, the disclosure relates to microelectronic devices

including local digit line structures and global digit line structures, and to related memory devices, and electronic systems.
BACKGROUND
(2) Microelectronic device designers often desire to increase the level of integration or density of features within a microelectronic device by reducing the dimensions of the individual features and by reducing the separation distance between neighboring features. In addition, microelectronic device designers often desire to design architectures that are not only compact, but offer performance advantages, as well as simplified, easier and less expensive to fabricate designs.
(3) One example of a microelectronic device is a memory device. Memory devices are generally provided as internal integrated circuits in computers or other electronic devices. There are many types of memory devices including, but not limited to, non-volatile memory devices (e.g., NAND Flash memory devices). One way of increasing memory density in non-volatile memory devices is to utilize vertical memory array (also referred to as a "three-dimensional (3D) memory array") architectures. A conventional vertical memory array includes strings of memory cells vertically extending through a stack structure including tiers of conductive structures and insulative materials. Each string of memory cells may include at least one select device coupled in series to a serial combination of vertically stacked memory cells. Such a configuration permits a greater number of switching devices (e.g., transistors) to be located in a unit of die area (e.g., length and width of active surface consumed) by building the array upwards (e.g., vertically) on a die, as compared to structures with conventional planar (e.g., two-dimensional (2D)) arrangements of transistors. However, conventional vertical memory array architectures may effectuate can hamper improvements in the performance (e.g., data transfer rates, power consumption) of the non-volatile memory device, and/or can impede reductions to the sizes (e.g., horizontal footprints) of features of the non-volatile memory device.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS
(1) FIGS. **1**A and **1**B are simplified, partial longitudinal cross-sectional (FIG. **1**A) and simplified, partial top-down (FIG. **1**B) views of a microelectronic device structure for a microelectronic device, in accordance with embodiments of the disclosure.
(2) FIG. **2** is a simplified, partial top-down view of a microelectronic device structures for a microelectronic device, in accordance with additional embodiments of the disclosure.
(3) FIG. **3** is simplified, partial longitudinal cross-sectional view of a microelectronic device structure for a microelectronic device, in accordance with further embodiments of the disclosure.
(4) FIGS. **4**A through **4**M are simplified, partial longitudinal cross-sectional views of a microelectronic device structure at different processing stages of a method of forming a microelectronic device, in accordance with embodiments of the disclosure.
(5) FIG. **5** is a schematic block diagram of an electronic system, in accordance with embodiments of the disclosure.
DETAILED DESCRIPTION
(6) The following description provides specific details, such as material compositions, shapes, and sizes, in order to provide a thorough description of embodiments of the disclosure. However, a person of ordinary skill in the art would understand that the embodiments of the disclosure may be practiced without employing these specific details. Indeed, the embodiments of the disclosure may be practiced in conjunction with conventional microelectronic device fabrication techniques employed in the industry. In addition, the description provided below does not form a complete process flow for manufacturing a microelectronic device (e.g., a memory device, such as 3D NAND Flash memory device). The structures described below do not form a complete

microelectronic device. Only those process acts and structures necessary to understand the embodiments of the disclosure are described in detail below. Additional acts to form a complete microelectronic device from the structures may be performed by conventional fabrication techniques.

(7) Drawings presented herein are for illustrative purposes only, and are not meant to be actual views of any particular material, component, structure, device, or system. Variations from the shapes depicted in the drawings as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments described herein are not to be construed as being limited to the particular shapes or regions as illustrated, but include deviations in shapes that result, for example, from manufacturing. For example, a region illustrated or described as box-shaped may have rough and/or nonlinear features, and a region illustrated or described as round may include some rough and/or linear features. Moreover, sharp angles that are illustrated may be rounded, and vice versa. Thus, the regions illustrated in the figures are schematic in nature, and their shapes are not intended to illustrate the precise shape of a region and do not limit the scope of the present claims. The drawings are not necessarily to scale. Additionally, elements common between figures may retain the same numerical designation.

(8) As used herein, a “memory device” means and includes microelectronic devices exhibiting memory functionality, but not necessary limited to memory functionality. Stated another way, and by way of non-limiting example only, the term “memory device” includes not only conventional memory (e.g., conventional non-volatile memory, such as conventional NAND memory; conventional volatile memory, such as conventional DRAM), but also includes an application specific integrated circuit (ASIC) (e.g., a system on a chip (SoC)), a microelectronic device combining logic and memory, and a graphics processing unit (GPU) incorporating memory.

(9) As used herein, the term “configured” refers to a size, shape, material composition, orientation, and arrangement of one or more of at least one structure and at least one apparatus facilitating operation of one or more of the structure and the apparatus in a pre-determined way.

(10) As used herein, the terms “vertical,” “longitudinal,” “horizontal,” and “lateral” are in reference to a major plane of a structure and are not necessarily defined by earth's gravitational field. A “horizontal” or “lateral” direction is a direction that is substantially parallel to the major plane of the structure, while a “vertical” or “longitudinal” direction is a direction that is substantially perpendicular to the major plane of the structure. The major plane of the structure is defined by a surface of the structure having a relatively large area compared to other surfaces of the structure. With reference to the figures, a “horizontal” or “lateral” direction may be perpendicular to an indicated “Z” axis, and may be parallel to an indicated “X” axis and/or parallel to an indicated “Y” axis; and a “vertical” or “longitudinal” direction may be parallel to an indicated “Z” axis, may be perpendicular to an indicated “X” axis, and may be perpendicular to an indicated “Y” axis.

(11) As used herein, features (e.g., regions, structures, devices) described as “neighboring” one another means and includes features of the disclosed identity (or identities) that are located most proximate (e.g., closest to) one another. Additional features (e.g., additional regions, additional structures, additional devices) not matching the disclosed identity (or identities) of the “neighboring” features may be disposed between the “neighboring” features. Put another way, the “neighboring” features may be positioned directly adjacent one another, such that no other feature intervenes between the “neighboring” features; or the “neighboring” features may be positioned indirectly adjacent one another, such that at least one feature having an identity other than that associated with at least one of the “neighboring” features is positioned between the “neighboring” features. Accordingly, features described as “vertically neighboring” one another means and includes features of the disclosed identity (or identities) that are located most vertically proximate (e.g., vertically closest to) one another. Moreover, features described as “horizontally neighboring” one another means and includes features of the disclosed identity (or identities) that are located most horizontally proximate (e.g., horizontally closest to) one another.

(12) As used herein, the term “intersection” means and includes a location at which two or more features (e.g., regions, structures, materials, devices) or, alternatively, two or more portions of a single feature meet. For example, an intersection between a first feature extending in a first direction (e.g., an X-direction) and a second feature extending in a second direction (e.g., a Y-direction) different than the first direction may be the location at which the first feature and the second feature meet.
(13) As used herein, spatially relative terms, such as “beneath,” “below,” “lower,” “bottom,” “above,” “upper,” “top,” “front,” “rear,” “left,” “right,” and the like, may be used for ease of description to describe one element's or feature's relationship to another element(s) or feature(s) as illustrated in the figures. Unless otherwise specified, the spatially relative terms are intended to encompass different orientations of the materials in addition to the orientation depicted in the figures. For example, if materials in the figures are inverted, elements described as “below” or “beneath” or “under” or “on bottom of” other elements or features would then be oriented “above” or “on top of” the other elements or features. Thus, the term “below” can encompass both an orientation of above and below, depending on the context in which the term is used, which will be evident to one of ordinary skill in the art. The materials may be otherwise oriented (e.g., rotated 90 degrees, inverted, flipped) and the spatially relative descriptors used herein interpreted accordingly.
(14) As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.
(15) As used herein, “and/or” includes any and all combinations of one or more of the associated listed items.
(16) As used herein, the phrase “coupled to” refers to structures operatively connected with each other, such as electrically connected through a direct Ohmic connection or through an indirect connection (e.g., by way of another structure).
(17) As used herein, the term “substantially” in reference to a given parameter, property, or condition means and includes to a degree that one of ordinary skill in the art would understand that the given parameter, property, or condition is met with a degree of variance, such as within acceptable tolerances. By way of example, depending on the particular parameter, property, or condition that is substantially met, the parameter, property, or condition may be at least 90.0 percent met, at least 95.0 percent met, at least 99.0 percent met, at least 99.9 percent met, or even 100.0 percent met.
(18) As used herein, “about” or “approximately” in reference to a numerical value for a particular parameter is inclusive of the numerical value and a degree of variance from the numerical value that one of ordinary skill in the art would understand is within acceptable tolerances for the particular parameter. For example, “about” or “approximately” in reference to a numerical value may include additional numerical values within a range of from 90.0 percent to 110.0 percent of the numerical value, such as within a range of from 95.0 percent to 105.0 percent of the numerical value, within a range of from 97.5 percent to 102.5 percent of the numerical value, within a range of from 99.0 percent to 101.0 percent of the numerical value, within a range of from 99.5 percent to 100.5 percent of the numerical value, or within a range of from 99.9 percent to 100.1 percent of the numerical value.
(19) As used herein, “conductive material” means and includes electrically conductive material such as one or more of a metal (e.g., tungsten (W), titanium (Ti), molybdenum (Mo), niobium (Nb), vanadium (V), hafnium (Hf), tantalum (Ta), chromium (Cr), zirconium (Zr), iron (Fe), ruthenium (Ru), osmium (Os), cobalt (Co), rhodium (Rh), iridium (Ir), nickel (Ni), palladium (Pa), platinum (Pt), copper (Cu), silver (Ag), gold (Au), aluminum (Al)), an alloy (e.g., a Co-based alloy, an Fe-based alloy, an Ni-based alloy, an Fe- and Ni-based alloy, a Co- and Ni-based alloy, an Fe- and Co-based alloy, a Co- and Ni- and Fe-based alloy, an Al-based alloy, a Cu-based alloy, a magnesium (Mg)-based alloy, a Ti-based alloy, a steel, a low-carbon steel, a stainless steel), a conductive metal-containing material (e.g., a conductive metal nitride, a conductive metal silicide, a conductive metal

carbide, a conductive metal oxide), and a conductively-doped semiconductor material (e.g., conductively-doped polysilicon, conductively-doped germanium (Ge), conductively-doped silicon germanium (SiGe)). In addition, a "conductive structure" means and includes a structure formed of and including conductive material.

(20) As used herein, "insulative material" means and includes electrically insulative material, such one or more of at least one dielectric oxide material (e.g., one or more of a silicon oxide (SiO.sub.x), phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass, an aluminum oxide (AlO.sub.x), a hafnium oxide (HfO.sub.x), a niobium oxide (NbO.sub.x), a titanium oxide (TiO.sub.x), a zirconium oxide (ZrO.sub.x), a tantalum oxide (TaO.sub.x), and a magnesium oxide (MgO.sub.x)), at least one dielectric nitride material (e.g., a silicon nitride (SiN.sub.y)), at least one dielectric oxynitride material (e.g., a silicon oxynitride (SiO.sub.xN.sub.y)), at least one dielectric oxycarbide material (e.g., silicon oxycarbide (SiO.sub.xC.sub.y)), at least one hydrogenated dielectric oxycarbide material (e.g., hydrogenated silicon oxycarbide (SiC.sub.xO.sub.yH.sub.z)), and at least one dielectric carboxynitride material (e.g., a silicon carboxynitride (SiO.sub.xC.sub.zN.sub.y)). Formulae including one or more of "x," "y," and "z" herein (e.g., SiO.sub.x, AlO.sub.x, HfO.sub.x, NbO.sub.x, TiO.sub.x, SiN.sub.y, SiO.sub.xN.sub.y, SiO.sub.xC.sub.y, SiC.sub.xO.sub.yH.sub.z, SiO.sub.xC.sub.zN.sub.y) represent a material that contains an average ratio of "x" atoms of one element, "y" atoms of another element, and "z" atoms of an additional element (if any) for every one atom of another element (e.g., Si, Al, Hf, Nb, Ti). As the formulae are representative of relative atomic ratios and not strict chemical structure, an insulative material may comprise one or more stoichiometric compounds and/or one or more non-stoichiometric compounds, and values of "x," "y," and "z" (if any) may be integers or may be non-integers. As used herein, the term "non-stoichiometric compound" means and includes a chemical compound with an elemental composition that cannot be represented by a ratio of well-defined natural numbers and is in violation of the law of definite proportions. In addition, an "insulative structure" means and includes a structure formed of and including insulative material.

(21) As used herein, the term "semiconductor material" refers to a material having an electrical conductivity between those of insulative materials and conductive materials. For example, a semiconductor material may have an electrical conductivity of between about 10.sup.−8 Siemens per centimeter (S/cm) and about 10.sup.4 S/cm (10.sup.6 S/m) at room temperature. Examples of semiconductor materials include elements found in column IV of the periodic table of elements such as silicon (Si), germanium (Ge), and carbon (C). Other examples of semiconductor materials include compound semiconductor materials such as binary compound semiconductor materials (e.g., gallium arsenide (GaAs)), ternary compound semiconductor materials (e.g., Al.sub.XGa.sub.1-XAs), and quaternary compound semiconductor materials (e.g., Ga.sub.XIn.sub.1-XAs.sub.YP.sub.1-Y), without limitation. Compound semiconductor materials may include combinations of elements from columns III and V of the periodic table of elements (III-V semiconductor materials) or from columns II and VI of the periodic table of elements (II-VI semiconductor materials), without limitation. Further examples of semiconductor materials include oxide semiconductor materials such as zinc tin oxide (Zn.sub.xSn.sub.yO, commonly referred to as "ZTO"), indium zinc oxide (In.sub.xZn.sub.yO, commonly referred to as "IZO"), zinc oxide (Zn.sub.xO), indium gallium zinc oxide (In.sub.xGa.sub.yZn.sub.zO, commonly referred to as "IGZO"), indium gallium silicon oxide (In.sub.xGa.sub.ySi.sub.zO, commonly referred to as "IGSO"), indium tungsten oxide (In.sub.xW.sub.yO, commonly referred to as "IWO"), indium oxide (In.sub.xO), tin oxide (Sn.sub.xO), titanium oxide (Ti.sub.xO), zinc oxide nitride (Zn.sub.xON.sub.z), magnesium zinc oxide (Mg.sub.xZn.sub.yO), zirconium indium zinc oxide (Zr.sub.xIn.sub.yZn.sub.zO), hafnium indium zinc oxide (Hf.sub.xIn.sub.yZn.sub.zO), tin indium zinc oxide (Sn.sub.xIn.sub.yZn.sub.zO), aluminum tin indium zinc oxide (Al.sub.xSn.sub.yIn.sub.zZn.sub.aO), silicon indium zinc oxide (Si.sub.xIn.sub.yZn.sub.zO),

aluminum zinc tin oxide (Al.sub.xZn.sub.ySn.sub.zO), gallium zinc tin oxide (Ga.sub.xZn.sub.ySn.sub.zO), zirconium zinc tin oxide (Zr.sub.xZn.sub.ySn.sub.zO), and other similar materials.

(22) As used herein, the term "homogeneous" means relative amounts of elements included in a feature (e.g., a region, a structures, a material) do not vary throughout different portions (e.g., different horizontal portions, different vertical portions) of the feature. Conversely, as used herein, the term "heterogeneous" means relative amounts of elements included in a feature (e.g., a material, a structure) vary throughout different portions of the feature. If a feature is heterogeneous, amounts of one or more elements included in the feature may vary stepwise (e.g., change abruptly), or may vary continuously (e.g., change progressively, such as linearly, parabolically) throughout different portions of the feature. The feature may, for example, be formed of and include a stack of at least two different materials.

(23) Unless the context indicates otherwise, the materials described herein may be formed by any suitable technique including, but not limited to, spin coating, blanket coating, chemical vapor deposition (CVD), plasma enhanced CVD (PECVD), atomic layer deposition (ALD), plasma enhanced ALD (PEALD), physical vapor deposition (PVD) (e.g., sputtering), or epitaxial growth. Depending on the specific material to be formed, the technique for depositing or growing the material may be selected by a person of ordinary skill in the art. In addition, unless the context indicates otherwise, removal of materials described herein may be accomplished by any suitable technique including, but not limited to, etching (e.g., dry etching, wet etching, vapor etching), ion milling, abrasive planarization (e.g., chemical-mechanical planarization (CMP)), or other known methods.

(24) FIG. **1**A is a simplified, partial longitudinal cross-sectional view of a microelectronic device structure **100** (e.g., a memory device structure, such as a 3D NAND Flash memory device structure) for a microelectronic device (e.g., a memory device, such as a 3D NAND Flash memory device), in accordance with embodiments of the disclosure. FIG. **1**B is simplified, partial top-down view of the microelectronic device structure **100** depicted in FIG. **1**A, wherein the view depicted in FIG. **1**A is about dashed line A.sub.1-A.sub.1 depicted in FIG. **1**B. For clarity and ease of understanding of the drawings and related description, not all features (e.g., regions, structures, materials, devices) of the microelectronic device structure **100** depicted in one of FIGS. **1**A and **1**B are depicted in the other of FIGS. **1**A through **1**B.

(25) Referring to FIG. **1**A, the microelectronic device structure **100** may be formed to include a base structure **102**; a stack structure **104** vertically overlying the base structure **102**, and including pillar structures **122** vertically extending therethrough; a select gate drain (SGD) plug tier **127** vertically overlying the stack structure **104**; a digit line tier **156** vertically overlying the SGD plug tier **127**; selector tiers **138** vertically interposed between the SGD plug tier **127** and the digit line tier **156**; and a conductive routing tier **150** vertically interposed between the digit line tier **156** and the selector tiers **138**. As described in further detail below, the microelectronic device structure **100** includes various features (e.g., regions, structures, materials, devices) individually operatively associated with (e.g., within; extending to, into, through, and/or between; physically and/or electrically connected to additional features of) one or more of the base structure **102**, the stack structure **104**, the SGD plug tier **127**, the digit line tier **156**, the selector tiers **138**, and the conductive routing tier **150**.

(26) The base structure **102** may comprise a base material or construction upon which additional features (e.g., materials, structures, devices) of the microelectronic device structure **100** are formed. The base structure **102** may comprise one or more of semiconductor material, conductive material, and insulative material. The base structure **102** may include an arrangement of different materials, different structures, and/or different regions. In some embodiments, the base structure **102** includes various circuitry (e.g., logic circuitry) therein.

(27) The stack structure **104** of the microelectronic device structure **100** may be formed to include a

vertically alternating sequence of conductive structures **106** and insulative structures **108** arranged in tiers **110**. The conductive structures **106** may be vertically interleaved with the insulative structures **108**. Each of the tiers **110** of the stack structure **104** may include at least one of the conductive structures **106** vertically neighboring at least one of the insulative structures **108**. The stack structure **104** may be formed to include any desired quantity of the tiers **110**, such as greater than or equal to sixteen (16) of the tiers **110**, greater than or equal to thirty-two (32) of the tiers **110**, greater than or equal to sixty-four (64) of the tiers **110**, greater than or equal to one hundred twenty-eight (128) of the tiers **110**, or greater than or equal to two hundred fifty-six (256) of the tiers **110**.
(28) The conductive structures **106** of the tiers **110** of the stack structure **104** may be formed of and include conductive material. By way of non-limiting example, the conductive structures **106** may each individually be formed of and include a metallic material comprising one or more of at least one metal, at least one alloy, and at least one conductive metal-containing material (e.g., a conductive metal nitride, a conductive metal silicide, a conductive metal carbide, a conductive metal oxide). In some embodiments, the conductive structures **106** are formed of and include W. Each of the conductive structures **106** may individually be substantially homogeneous, or one or more of the conductive structures **106** may individually be substantially heterogeneous. In some embodiments, each of the conductive structures **106** is formed to be substantially homogeneous.
(29) Optionally, one or more liner materials(s) (e.g., insulative liner material(s), conductive liner material(s)) may also be formed around the conductive structures **106**. The liner material(s) may, for example, be formed of and include one or more a metal (e.g., titanium, tantalum), an alloy, a metal nitride (e.g., tungsten nitride, titanium nitride, tantalum nitride), and a metal oxide (e.g., aluminum oxide). In some embodiments, the liner material(s) comprise at least one conductive material employed as a seed material for the formation of the conductive structures **106**. In some embodiments, the liner material(s) comprise titanium nitride. In further embodiments, the liner material(s) further include aluminum oxide. As a non-limiting example, aluminum oxide may be formed directly adjacent the insulative structures **108**, titanium nitride may be formed directly adjacent the aluminum oxide, and tungsten may be formed directly adjacent the titanium nitride. For clarity and ease of understanding the description, the liner material(s) are not illustrated in FIGS. **1**A and **1**B, but it will be understood that the liner material(s) may be disposed around the conductive structures **106**.
(30) The insulative structures **108** of the tiers **110** of the stack structure **104** may be formed of and include insulative material, such one or more of at least one dielectric oxide material (e.g., one or more of SiO.sub.x, phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass, AlO.sub.x, HfO.sub.x, NbO.sub.x, TiO.sub.x, ZrO.sub.x, TaO.sub.x, and MgO.sub.x), at least one dielectric nitride material (e.g., SiN.sub.y), at least one dielectric oxynitride material (e.g., SiO.sub.xN.sub.y), and at least one dielectric carboxynitride material (e.g., SiO.sub.xC.sub.zN.sub.y). In some embodiments, each of the insulative structures **108** is formed of and includes a dielectric oxide material, such as SiO.sub.x (e.g., SiO.sub.2). Each of the insulative structures **108** may individually be substantially homogeneous, may be substantially heterogeneous. In some embodiments, each of the insulative structures **108** is substantially homogeneous.
(31) Referring to FIG. **1**B, the stack structure **104** may be divided (e.g., separated, partitioned) into blocks **164** separated from one another by dielectric slot structures **166** (e.g., dielectric-filled slots, dielectric-filled openings). The dielectric slot structures **166** may vertically extend (e.g., in the Z-direction) completely through the stack structure **104**. The blocks **164** of the stack structure **104** may be formed to horizontally extend parallel in an X-direction. As used herein, the term “parallel” means substantially parallel. Horizontally neighboring blocks **164** of the stack structure **104** may be separated from one another in a Y-direction orthogonal to the X-direction by the dielectric slot structures **166**. The dielectric slot structures **166** may also horizontally extend parallel in the X-direction. Each of the blocks **164** of the stack structure **104** may exhibit substantially the same

geometric configuration (e.g., substantially the same dimensions and substantially the same shape) as each other of the blocks **164**, or one or more of the blocks **164** may exhibit a different geometric configuration (e.g., one or more different dimensions and/or a different shape) than one or more other of the blocks **164**. In addition, each pair of horizontally neighboring blocks **164** of the stack structure **104** may be horizontally separated from one another by substantially the same distance (e.g., corresponding to a width in the Y-direction of each of the dielectric slot structures **166**) as each other pair of horizontally neighboring blocks **164** of the stack structure **104**, or at least one pair of horizontally neighboring blocks **164** of the stack structure **104** may be horizontally separated from one another by a different distance than that separating at least one other pair of horizontally neighboring blocks **164** of the stack structure **104**. In some embodiments, the blocks **164** of the stack structure **104** are substantially uniformly (e.g., substantially non-variably, substantially equally, substantially consistently) sized, shaped, and spaced relative to one another.

(32) Still referring to FIG. **1**B, each block **164** of the stack structure **104** may be sub-divided into multiple (e.g., a plurality of, more than one) sub-blocks **114**. The sub-blocks **114** of an individual block **164** may horizontally extend parallel in the X-direction. Each of the sub-blocks **114** may individually be operatively associated with at least one row of the pillar structures **122** extending the X-direction. For example, as depicted in FIG. **1**B, an individual block **164** of the stack structure **104** may include sixteen (16) sub-blocks **114** operatively associated with sixteen (16) rows of the pillar structures **122**. In addition, for an individual block **164**, multiple (e.g., a plurality of, more than one) sub-blocks **114** thereof may be grouped together with one another within sub-block groups **116**. As depicted in FIG. **1**B, an individual block **164** of the stack structure **104** may, for example, include four (4) sub-block groups **116** each including four (4) of the sub-blocks **114**. By way of non-limiting example, the sub-block groups **116** may include a first sub-block group **116**A, a second sub-block group **116**B, a third sub-block group **116**C, and a fourth sub-block group **116**D; and each of the sub-block groups **116** may individually include a first sub-block **114**A, a second sub-block **114**B, a third sub-block **114**C, and a fourth sub-block **114**D. Sub-block groups **116** horizontally neighboring one another in the Y-direction within an individual block **164** may exhibit an inverse horizontal arrangement of the different sub-blocks **114** thereof relative to one another. For example, the fourth sub-block **114**D of the first sub-block group **116**A may be most horizontally proximate the fourth sub-block **114**D of the second sub-block group **116**B horizontally neighboring the first sub-block group **116**A; the first sub-block **114**A of the second sub-block group **116**B may be most horizontally proximate to the first sub-block **114**A of the third sub-block group **116**C horizontally neighboring the second sub-block group **116**B; and the fourth sub-block **114**D of the third sub-block group **116**C may be most horizontally proximate to the fourth sub-block **114**D of the fourth sub-block group **116**D horizontally neighboring the third sub-block group **116**C.

(33) While FIG. **1**B depicts an individual block **164** of the stack structure **104** as including sixteen (16) sub-blocks **114** and four (4) sub-block groups **116** each including a different four (4) of the sixteen (16) sub-blocks **114**, in additional embodiments one or more (e.g., each) of the blocks **164** of the stack structure **104** includes a different quantity of sub-blocks **114**, a different quantity of sub-block groups **116**, and/or a different quantity of sub-blocks **114** within an individual sub-block group **116** thereof. For example, an individual block **164** may include greater than sixteen (16) sub-blocks **114** or less than sixteen (16) sub-blocks **114**; may include greater than four (4) sub-block groups **116** or less four (4) sub-block groups **116**; and/or may include greater than four (4) sub-blocks **114** in an individual sub-block group **116** or less than four (4) sub-blocks **114** in an individual sub-block group **116**. For an individual block **164** of the stack structure **104**, the quantity of sub-blocks **114**, the quantity of sub-block groups **116**, and the quantity of sub-blocks **114** per sub-block group **116** may be selected, at least in part, based on the horizontal area of the block **164** as well as the horizontal areas and horizontal positions of the pillar structures **122** located within the block **164**.

(34) Referring collectively to FIGS. **1**A and **1**B, it will be recognized that the dashed line A.sub.1-

A.sub.1 depicted in FIG. **1**B, which identifies the position and orientation of the simplified, partial longitudinal cross-sectional view of the microelectronic device structure **100** shown in FIG. **1**A, horizontally extends diagonal to each of the X-direction (e.g., first horizontal direction) and the Y-direction (e.g., second horizontal direction orthogonal to the first horizontal direction) shown in FIG. **1**A. Put another way, the horizontal orientation of the dashed line A.sub.1-A.sub.1 depicted in FIG. **1**B (and, hence, the longitudinal cross-sectional view of FIG. **1**A), is acutely angled relative to each of the X-direction and the Y-direction shown in FIG. **1**B. Accordingly, it will be appreciated that the horizontal orientations some features (e.g., some structures, some devices, some regions) of the microelectronic device structure **100** depicted in FIG. **1**A and described in further details below, such as (but without limitation) some features within a sub-section B.sub.1 shown in FIGS. **1**A and **1**B, are also acutely angled relative to each of the X-direction and the Y-direction shown in FIG. **1**B.

(35) Referring to FIG. **1**A, the tiers **110** of the stack structure **104** may be grouped into different tier sections **111**. The tier sections **111** may include an access line tier section **111**A, a stacked select gate drain (SGD) tier section **111**B overlying the access line tier section **111**A, and a sense node tier section **111**C overlying the stacked SGD tier section **111**B. In addition, the tier sections **111** of the stack structure **104** may further include a source side select (SGS) gate tier section **111**D underlying the access line tier section **111**A. As described in further details below, during use and operation of a microelectronic device including the microelectronic device structure **100**, at least some of the tiers **110** within the stacked SGD tier section **111**B may be employed to select different sub-blocks **114** within individual blocks **164** (FIG. **1**B) of the stack structure **104**; and at least some other of the tiers **110** within the sense node tier section **111**C may be employed for capacitive-sense operations (amongst other operations) for the individual blocks **164** (FIG. **1**B) of the stack structure **104**.

(36) As shown in FIG. **1**A, a first group of the tiers **110** of the stack structure **104** within the access line tier section **111**A may include an active access line tier **110**A, and a dummy access line tier **110**B overlying the active access line tier **110**A. For an individual block **164** (FIG. **1**B) of the stack structure **104**, a conductive structure **106** of the active access line tier **110**A may be employed as a so-called “active” access line structure facilitating electrical communication between two or more components (e.g., memory cells, string drivers) of a microelectronic device including the microelectronic device structure **100**; and a conductive structure **106** of the dummy access line tier **110**B may be employed as a so-called “dummy” access line structure that does not facilitate electrical communication between two or more components (e.g., memory cells, string drivers) of a microelectronic device including the microelectronic device structure **100**. While FIG. **1**A only depicts the access line tier section **111**A as including one (1) active access line tier **110**A and one (1) dummy access line tier **110**B the disclosure is not so limited. Rather, the access line tier section **111**A may include greater than one active access line tier **110**A (e.g., greater than or equal to eight (8) active access line tiers **110**A, greater than or equal to sixteen (16) active access line tiers **110**A, greater than or equal to thirty-two (32) active access line tiers **110**A, greater than or equal to sixty-four (64) active access line tiers **110**A, greater than or equal to one-hundred twenty-eight (128) active access line tiers **110**A, greater than or equal to two-hundred fifty-six (256) active access line tiers **110**A); and/or greater than one dummy access line tier **110**B (e.g., greater than or equal to two (2) dummy access line tiers **110**B).

(37) Still referring to FIG. **1**A, a second group of the tiers **110** of the stack structure **104** within the stacked SGD tier section **111**B may include a program-inhibit SGD tier **110**C overlying the dummy access line tier **110**B, a first SGD bar tier **110**D (e.g., a first “complementary” SGD tier) overlying the program-inhibit SGD tier **110**C, a second SGD bar tier **110**E (e.g., a second “complementary” SGD tier) overlying the first SGD bar tier **110**D, a first SGD tier **110**F (e.g., a first “true” SGD tier) overlying the second SGD bar tier **110**E, a second SGD tier **110**G (e.g., a second “true” SGD tier) overlying the first SGD tier **110**F, and a read-amplification SGD tier **110**H overlying the second

SGD tier **110**G. For an individual block **164** (FIG. **1**B) of the stack structure **104**, a conductive structure **106** of the program-inhibit SGD tier **110**C may be employed as a program-inhibit SGD structure; a conductive structure **106** of the first SGD bar tier **110**D may be employed as a first SGD bar structure; a conductive structure **106** of the second SGD bar tier **110**E may be employed as a second SGD bar structure; a conductive structure **106** of the first SGD tier **110**F may be employed as a first SGD structure; a conductive structure **106** of the second SGD tier **110**G may be employed as a second SGD structure; and a conductive structure **106** of the read-amplification SGD tier **110**H may be employed as a read-amplification SGD structure.

(38) With continued reference to FIG. **1**A, a third group of the tiers **110** of the stack structure **104** within the sense node tier section **111**C may include a first select gate programming (SGP) bar tier **110**I (e.g., a first “complementary” SGP tier) overlying the read-amplification SGD tier **110**H, a second SGP bar tier **110**J (e.g., a second “complementary” SGP tier) overlying the first SGP bar tier **110**I, a first SGP tier **110**K (e.g., a first “true” SGP tier) overlying the second SGP bar tier **110**J, a second SGP tier **110**L (e.g., a second “true” SGP tier) overlying the first SGP tier **110**K, and a gate-induced drain-leakage (GIDL) generation (GG) tier **110**M overlying the second SGP tier **110**L. For an individual block **164** (FIG. **1**B) of the stack structure **104**, a conductive structure **106** of the first SGP bar tier **110**I may be employed as a first SGP bar structure; a conductive structure **106** of the second SGP bar tier **110**J may be employed as a second SGD bar structure; a conductive structure **106** of the first SGP tier **110**K may be employed as a first SGP structure; a conductive structure **106** of the second SGP tier **110**L may be employed as a second SGP structure; and a conductive structure **106** of the GG tier **110**M may be employed as a GG structure. As described in further detail below, portions of different tiers **110** (e.g., the first SGP bar tier **110**I, the second SGP bar tier **110**J, the first SGP tier **110**K, the second SGP tier **110**L) within the stacked SGD tier section **111**B may be doped with one or more conductivity-enhancing species (e.g., N-type dopant, P-type dopant) that permit horizontally neighboring select transistors within vertical boundaries of an individual tier **110** of the sense node tier section **111**C to have different threshold voltage (V.sub.t) properties than one another. The third group of the tiers **110** of the stack structure **104** within the sense node tier section **111**C facilitate associating different pillar structures **122** vertically extending through an individual block **164** (FIG. **1**B) with different sub-blocks **114** of the block **164** (FIG. **1**B) by programming selected transistors at vertical elevations of different tiers **110** within the sense node tier section **111**C to desired V.sub.t levels, as described in further detail below.

(39) Optionally, one or more additional “dummy” tiers (e.g., in additional to the dummy access line tiers **110**B) may be included within the stack structure **104**, vertically between two or more other of the tiers **110**. A conductive structure **106** of an individual additional dummy tier may be employed as an additional so-called “dummy” structure that does not facilitate electrical communication between two or more components of a microelectronic device including the microelectronic device structure **100**. If included, additional dummy tier(s) may be utilized to mitigate so-called “trap-up” and “downshift” of V.sub.t by relaxing the electric field between devices in ON and OFF state. An additional dummy tier may be biased at intermediate potential to mitigate (e.g., avoid) a too abrupt potential difference between the tiers at HIGH and LOW biases. As a non-limiting example, an additional dummy tier may be vertically interposed between the program-inhibit SGD tier **110**C and the first SGD bar tier **110**D. As another non-limiting example, an additional dummy tier may be vertically interposed between the second SGD tier **110**G and the read-amplification SGD tier **110**H. As a further non-limiting example, an additional dummy tier may be vertically interposed between the read-amplification SGD tier **110**H and the first SGP bar tier **110**I.

(40) Still referring to FIG. **1**A, the pillar structures **122** may vertically extend completely through the tiers **110** of the stack structure **104**. For example, the pillar structures **122** may individually extend through each of the sense node tier section **111**C, the stacked SGD tier section **111**B, the access line tier section **111**A, and the SGS gate tier section **111**D, and to or into the base structure **102**. As described in further detail below, the pillar structures **122** may each individually be formed

of and include a stack of materials. By way of non-limiting example, each of the pillar structures **122** may be formed to include a charge-blocking material, such as first dielectric oxide material (e.g., SiO.sub.x, such as SiO.sub.2; AlO.sub.x, such as Al.sub.2O.sub.3); a charge-trapping material, such as a dielectric nitride material (e.g., SiN.sub.y, such as Si.sub.3N.sub.4); a gate dielectric material, such as a second dielectric oxide material (e.g., SiO.sub.x, such as SiO.sub.2); a channel material, such as a semiconductor material (e.g., silicon, such as polycrystalline silicon); and a dielectric fill material (e.g., a dielectric oxide, a dielectric nitride, air). The charge-blocking material may be formed on or over surfaces of the conductive structures **106** and the insulative structures **108** of the tiers **110** of stack structure **104** at least partially defining horizontal boundaries of the pillar structures **122**; the charge-trapping material may be horizontally surrounded by the charge-blocking material; the gate dielectric material may be horizontally surrounded by the charge-trapping material; the channel material may be horizontally surrounded by the gate dielectric material; and the dielectric fill material may be horizontally surrounded by the channel material.

(41) The pillar structures **122** may individually exhibit a desirable geometric configuration (e.g., dimensions, shape), and may also be distributed relative to one another in a desirable manner within horizontal areas of the blocks **164** (FIG. **1**B), of the stack structure **104**. The pillar structures **122** may individually exhibit a critical dimension (CD) (e.g., maximum horizontal dimension) less than about 120 nanometers (nm), such as less than or equal to about 110 nm, less than or equal to about 100 nm, less than or equal to about 90 nm, or less than or equal to about 80 nm. In some embodiments, the pillar structures **122** individually exhibit a CD within a range of from about 70 nm to about 80 nm. A pitch between pillar structures **122** horizontally neighboring one another may be less than about 140 nm, such as less than or equal to about 130 nm, less than or equal to about 120 nm, less than or equal to about 110 nm, or less than or equal to about 100 nm. In some embodiments, a pitch between pillar structures **122** horizontally neighboring one another is within a range of from about 90 nm to about 100 nm. A taper of each pillar structure **122**, as defined by the difference between a largest horizontal dimension of the pillar structure **122** (e.g., at an upper vertical boundary thereof) and a smallest horizontal dimension of the pillar structure **122** (e.g., at a lower vertical boundary thereof), may be less than or equal to about 20 nm, such as less than or equal to about 19 nm, or less than or equal to about 18 nm. In some embodiments, a taper of each pillar structure **122** is within a range of from about 16 nm to about 18 nm. In addition, as shown in FIG. **1**B, in some embodiments, each block **164** of the stack structure **104** exhibits a hexagonal distribution (e.g., a hexagonal pattern) of the pillar structures **122**. As described in further detail below, the horizontal dimensions and pitch of the pillar structures **122** may respectively be relatively smaller than those for conventional pillar structures as a result of lower sense current (I sense) requirements facilitated by the configuration of the microelectronic device structure **100** relative to conventional configurations.

(42) Referring to FIG. **1**A, intersections of the pillar structures **122** and the conductive structures **106** of the active access line tiers (e.g., the active access line tier **110**A) within the access line tier section **111**A of the stack structure **104** may define vertically extending strings of memory cells **121** coupled in series with one another within the stack structure **104**. As shown in FIG. **1**A, intersections of the pillar structures **122** and the conductive structure **106** of the active access line tier **110**A may define memory cells **121** at the horizontal positions of the pillar structures **122** and at the vertical position of the conductive structure **106**. Intersections of the pillar structures **122** and the conductive structures **106** of additional active access line tiers vertically underling the active access line tier **110**A may define more of the memory cells **121** at the horizontal positions of the pillar structures **122** and at the vertical positions of the conductive structures **106** of the additional active access line tiers. In some embodiments, the memory cells **121** formed at the intersections of the conductive structures **106** of the active access line tiers (e.g., the active access line tier **110**A) and the pillar structures **122** within access line tier section **111**A comprise so-called “MONOS”

(metal-oxide-nitride-oxide-semiconductor) memory cells. In additional embodiments, the memory cells **121** comprise so-called “TANOS” (tantalum nitride-aluminum oxide-nitride-oxide-semiconductor) memory cells, or so-called “BETANOS” (band/barrier engineered TANOS) memory cells, each of which are subsets of MONOS memory cells. In further embodiments, the memory cells **121** comprise so-called “floating gate” memory cells including floating gates (e.g., metallic floating gates) as charge storage structures. The floating gates may horizontally intervene between central structures of the pillar structures **122** and the conductive structures **106** of the different active access line tiers (e.g., the active access line tier **110**A) of the stack structure **104**. The vertically extending strings of memory cells **121** together form at least one memory array within the stack structure **104**.

(43) Intersections of the pillar structures **122** and the conductive structures **106** within the tiers **110** of the sense node tier section **111**C, the stacked SGD tier section **111**B, and the SGS gate tier section **111**D of the stack structure **104** may define different select transistors **123** coupled in series with the vertically extending strings of memory cells **121**. As described in further detail below, the select transistors **123** within the sense node tier section **111**C, the stacked SGD tier section **111**B, and the SGS gate tier section **111**D of the stack structure **104** may be employed for various memory array operations (e.g., program operations, erase operations, read operations) during use and operation of a microelectronic device including the microelectronic device structure **100**.

(44) Still referring to FIG. **1**A, the microelectronic device structure **100** includes different threshold voltage (V.sub.t)-enhanced sections **118** at different vertical and horizontal positions therein. By way of non-limiting example, within horizontal areas of the blocks **164** (FIG. **1**B), the microelectronic device structure **100** may include V.sub.t-enhanced sections **118** at vertical positions of at least some of the tiers **110** of the stack structure **104** within the stacked SGD tier section **111**B and the sense node tier section **111**C. Some of the V.sub.t-enhanced sections **118** may include portions of a tier **110** of the stack structure **104** doped with at least one conductivity-enhancing species (e.g., at least one P-type dopant), as well as portions of one or more pillar structures **122** doped with the at least one conductivity-enhancing species. Some other of the V.sub.t-enhanced sections **118** may include portions of a tier **110** of the stack structure **104** not doped with at least one conductivity-enhancing species, and portions of one or more pillar structures **122** also not doped with the at least one conductivity-enhancing species. Even in the absence of the conductivity-enhancing species within boundaries thereof, an individual V.sub.t-enhanced section **118** may be considered to be “V.sub.t-enhanced” as a result of device (e.g., transistor) V.sub.t characteristics facilitated within the boundaries thereof by way of configurations of additional features (e.g., SGD plug structures) of the microelectronic device structure **100** in conjunction with programming operations performed for a microelectronic device including microelectronic device structure **100**, as described in further detail below. For example, a select transistor **123** associated with a portions of pillar structure **122** within vertical and horizontal boundaries of an individual V.sub.t-enhanced section **118** substantially free of conductivity-enhancing species may be provided with enhanced V.sub.t characteristics due, in part, to a material composition (e.g., N-type doped semiconductive material) of an SGD plug structure coupled to the pillar structure **122**, as described in further detail below. Within a horizontal area of an individual block **164** (FIG. **1**B) of the stack structure **104**, each V.sub.t-enhanced section **118** may individually vertically overlap at least the conductive structure **106** of one (1) of the tiers **110** of the stack structure **104**, and may encompass at least two (2) horizontally neighboring pillar structures **122** within different sub-blocks **114** than one another. An individual V.sub.t-enhanced section **118** may be substantially confined within horizontal boundaries of an individual sub-block group **116** of the block **164** (FIG. **1**B), or may horizontally extend at least partially across and between at least two (2) horizontally neighboring sub-block groups **116** of the block **164** (FIG. **1**B).

(45) The V.sub.t-enhanced sections **118** of the microelectronic device structure **100** may be used to provide different select transistors **123** within the stacked SGD tier section **111**B and the sense node

tier section **111**C of the stack structure **104** with desired V.sub.t characteristics. For example, within an individual block **164** (FIG. **1**B) of the stack structure **104**, the V.sub.t-enhanced sections **118** may be configured such that some horizontally neighboring select transistors **123** substantially vertically aligned with one another (e.g., within the same tier **110** as one another) and operatively associated with pillar structures **122** in different sub-blocks **114** of the block **164** (FIG. **1**B) than one another have different V.sub.t characteristics (e.g., are set to different V.sub.t levels) than one another; and such that some other horizontally neighboring select transistors **123** substantially vertically aligned with one another and operatively associated with pillar structures **122** in different sub-blocks **114** of the block **164** (FIG. **1**B) than one another have substantially the same V.sub.t characteristics (e.g., are set to the same V.sub.t levels) as one another. As another example, within an individual block **164** (FIG. **1**B) of the stack structure **104**, the V.sub.t-enhanced sections **118** may be configured such that some vertically neighboring select transistors **123** operatively associated with the same pillar structure **122** as one another have substantially the same V.sub.t characteristics (e.g., are set to the same V.sub.t levels) as one another; and such that such that some other vertically neighboring select transistors **123** operatively associated with the same pillar structure **122** as one another have different V.sub.t properties (e.g., are set to different V.sub.t levels) than one another. As an additional example, within an individual block **164** (FIG. **1**B) of the stack structure **104**, the V.sub.t-enhanced sections **118** may be configured such that some select transistors **123** vertically and horizontally neighboring one another and operatively associated with pillar structures **122** in different sub-blocks **114** of the block **164** (FIG. **1**B) than one another have different V.sub.t properties (e.g., are set to different V.sub.t levels) than one another; and such that some other select transistors **123** vertically and horizontally neighboring one another and operatively associated with pillar structures **122** in different sub-blocks **114** of the block **164** (FIG. **1**B) than one another have substantially the same V.sub.t properties (e.g., are set to the same V.sub.t levels) as one another.

(46) As shown in FIG. **1**A, in some embodiments, the V.sub.t-enhanced sections **118** of the microelectronic device structure **100** include first V.sub.t-enhanced sections **118**A at a vertical elevation of the program-inhibit SGD tier **110**C; second V.sub.t-enhanced sections **118**B at a vertical elevation of the first SGD bar tier **110**D; third V.sub.t-enhanced sections **118**C at a vertical elevation the second SGD bar tier **110**E; fourth V.sub.t-enhanced sections **118**D at a vertical elevation of the first SGD tier **110**F; fifth V.sub.t-enhanced sections **118**E at a vertical elevation of the second SGD tier **110**G; sixth V.sub.t-enhanced sections **118**E at a vertical elevation of the read-amplification SGD tier **110**H; seventh V.sub.t-enhanced sections **118**G at a vertical elevation of the first SGP bar tier **110**I; eighth V.sub.t-enhanced sections **118**H at a vertical elevation of the second SGP bar tier **110**J; ninth V.sub.t-enhanced sections **118**I at a vertical elevation of the first SGP tier **110**K; and tenth V.sub.t-enhanced sections **118**J at a vertical elevation of the second SGP tier **110**L. Within an individual sub-block group **116** of an individual block **164** (FIG. **1**B), different V.sub.t-enhanced sections **118** may encompass different sub-blocks **114** of the sub-block group **116**. Non-limiting examples of configurations of some of the V.sub.t-enhanced sections **118** within a horizontal area of the sub-section B.sub.1 of the microelectronic device structure **100** shown in FIGS. **1**A and **1**B are described in further detail below.

(47) Referring to FIG. **1**A, within a horizontal area of the sub-section B.sub.1 of the microelectronic device structure **100**, one first V.sub.t-enhanced section **118**A may encompass portions of the first sub-block **114**A, the second sub-block **114**B, the third sub-block **114**C, and the fourth sub-block **114**D of the third sub-block group **116**C. The first V.sub.t-enhanced section **118**A may comprise portions of the program-inhibit SGD tier **110**C, as well as portions of four (4) of the pillar structures **122** at the vertical position of the program-inhibit SGD tier **110**C. The four (4) of the pillar structures **122** may include a first pillar structure **122**A, a second pillar structure **122**B, a third pillar structure **122**C, and a fourth pillar structure **122**D. The four (4) of the pillar structures **122** may be included in one (1) of multiple pillar groups **124** of within horizontal boundaries of the

third sub-block group **116**C, wherein each of the pillar groups **124** individually includes four (4) pillar structures **122** that are different than four (4) other pillar structures **122** of each other of the pillar groups **124**. Each of the four (4) of the pillar structures **122** within the horizontal area of the sub-section B.sub.1 may be positioned in a different one of the first sub-block **114**A, the second sub-block **114**B, the third sub-block **114**C, and the fourth sub-block **114**D of the third sub-block group **116**C than each other of the four (4) of the pillar structures **122**. The first V.sub.t-enhanced section **118**A may be substantially free of conductivity-enhancing species (e.g., P-type dopants, N-type dopants) therein. Select transistors **123** within the first V.sub.t-enhanced section **118**A may be configured (e.g., programmed) to have substantially the same V.sub.t characteristics as one another.

(48) Within the horizontal area of the sub-section B **1** of the microelectronic device structure **100**, one second V.sub.t-enhanced section **118**B may encompass portions of the third sub-block **114**C and the fourth sub-block **114**D of the third sub-block group **116**C. The second V.sub.t-enhanced section **118**B may not encompass portions of either of the first sub-block **114**A and the second sub-block **114**B of the third sub-block group **116**C. The second V.sub.t-enhanced section **118**B may comprise portions of the first SGD bar tier **110**D, as well as portions of two (2) of the pillar structures **122** of the pillar group **124** at the vertical position of the first SGD bar tier **110**D. In some embodiments, the two (2) of the pillar structures **122** include the third pillar structure **122**C and the fourth pillar structure **122**D. The second V.sub.t-enhanced section **118**B may not include portions of the other two (2) of the pillar structures **122** of the pillar group **124**. In some embodiments, the other two (2) of the pillar structures **122** include the first pillar structure **122**A and the second pillar structure **122**B. The second V.sub.t-enhanced section **118**B may be substantially free of conductivity-enhancing species (e.g., P-type dopants, N-type dopants) therein. Select transistors **123** within the second V.sub.t-enhanced section **118**B may be configured (e.g., programmed) to have substantially the same V.sub.t characteristics as one another.

(49) Within the horizontal area of the sub-section B.sub.1 of the microelectronic device structure **100**, one third V.sub.t-enhanced section **118**C may encompass portions of the second sub-block **114**B and the third sub-block **114**C of the third sub-block group **116**C. The third V.sub.t-enhanced section **118**C may not encompass portions of either of the first sub-block **114**A and the fourth sub-block **114**D of the third sub-block group **116**C. The third V.sub.t-enhanced section **118**C may comprise portions of the second SGD bar tier **110**E doped with at least one conductivity-enhancing species, as well as portions of two (2) of the pillar structures **122** of the pillar group **124** at the vertical position of the second SGD bar tier **110**E. In some embodiments, the two (2) of the pillar structures **122** include the second pillar structure **122**B and the third pillar structure **122**C. The third V.sub.t-enhanced section **118**C may not include portions of the other two (2) of the pillar structures **122** of the pillar group **124**. In some embodiments, the other two (2) of the pillar structures **122** include the first pillar structure **122**A and the fourth pillar structure **122**D. The third V.sub.t-enhanced section **118**C may be substantially free of conductivity-enhancing species (e.g., P-type dopants, N-type dopants) therein. Select transistors **123** within the third V.sub.t-enhanced section **118**C may be configured (e.g., programmed) to have substantially the same V.sub.t characteristics as one another.

(50) Within the horizontal area of the sub-section B.sub.1 of the microelectronic device structure **100**, one fourth V.sub.t-enhanced section **118**D may encompass portions of the first sub-block **114**A and the second sub-block **114**B of the third sub-block group **116**C. The fourth V.sub.t-enhanced section **118**D may not encompass portions of either of the third sub-block **114**C and the fourth sub-block **114**D of the third sub-block group **116**C. The fourth V.sub.t-enhanced section **118**D may comprise portions of the first SGD tier **110**F doped with at least one conductivity-enhancing species, as well as portions of two (2) of the pillar structures **122** of the pillar group **124** at the vertical position of the first SGD tier **110**F. In some embodiments, the two (2) of the pillar structures **122** include the first pillar structure **122**A and the second pillar structure **122**B. The fourth V.sub.t-enhanced section **118**D may not include portions of the other two (2) of the pillar

structures **122** of the pillar group **124**. In some embodiments, the other two (2) of the pillar structures **122** include the third pillar structure **122**C and the fourth pillar structure **122**D. The fourth V.sub.t-enhanced section **118**D may be substantially free of conductivity-enhancing species (e.g., P-type dopants, N-type dopants) therein. Select transistors **123** within the fourth V.sub.t-enhanced section **118**D may be configured (e.g., programmed) to have substantially the same V.sub.t characteristics as one another.

(51) Within the horizontal area of the sub-section B.sub.1 of the microelectronic device structure **100**, one fifth V.sub.t-enhanced section **118**E may encompass a portion of the first sub-block **114**A and another fifth V.sub.t-enhanced section **118**E may encompass a portion of the fourth sub-block **114**D. Neither of the fifth V.sub.t-enhanced sections **118**E may encompass portions of either of the second sub-block **114**B and the third sub-block **114**C of the third sub-block group **116**C. The fifth V.sub.t-enhanced sections **118**E may comprise portions of the second SGD tier **110**G doped with at least one conductivity-enhancing species, as well as portions of two (2) of the pillar structures **122** of the pillar group **124** at the vertical position of the second SGD tier **110**G. In some embodiments, the two (2) of the pillar structures **122** include the first pillar structure **122**A and the fourth pillar structure **122**D. The fifth V.sub.t-enhanced sections **118**E may not include portions of the other two (2) of the pillar structures **122** of the pillar group **124**. In some embodiments, the other two (2) of the pillar structures **122** include the second pillar structure **122**B and the third pillar structure **122**C. The fifth V.sub.t-enhanced sections **118**E may be substantially free of conductivity-enhancing species (e.g., P-type dopants, N-type dopants) therein. Select transistors **123** within the fifth V.sub.t-enhanced sections **118**E may be configured (e.g., programmed) to have substantially the same V.sub.t characteristics as one another.

(52) Within the horizontal area of the sub-section B.sub.1 of the microelectronic device structure **100**, one sixth V.sub.t-enhanced section **118**F may encompass portions of the first sub-block **114**A, the second sub-block **114**B, the third sub-block **114**C, and the fourth sub-block **114**D of the third sub-block group **116**C. The sixth V.sub.t-enhanced section **118**F may comprise portions of the read-amplification SGD tier **110**H doped with at least one conductivity-enhancing species, as well as portions of all four (4) of the pillar structures **122** of the pillar structures **122** of the pillar group **124** at the vertical position of the read-amplification SGD tier **110**H. The sixth V.sub.t-enhanced section **118**F may be substantially free of conductivity-enhancing species (e.g., P-type dopants, N-type dopants) therein. Select transistors **123** within the sixth V.sub.t-enhanced section **118**F may be configured (e.g., programmed) to have substantially the same V.sub.t characteristics as one another.

(53) Within the horizontal area of the sub-section B.sub.1 of the microelectronic device structure **100**, one seventh V.sub.t-enhanced section **118**G may encompass portions of the third sub-block **114**C and the fourth sub-block **114**D of the third sub-block group **116**C. The seventh V.sub.t-enhanced section **118**G may not encompass portions of either of the first sub-block **114**A and the second sub-block **114**B of the third sub-block group **116**C. The seventh V.sub.t-enhanced section **118**G may comprise portions of the first SGP bar tier **110**I doped with at least one conductivity-enhancing species, as well as portions of two (2) of the pillar structures **122** of the pillar group **124** at the vertical position of the first SGP bar tier **110**I. In some embodiments, the two (2) of the pillar structures **122** include the third pillar structure **122**C and the fourth pillar structure **122**D. The seventh V.sub.t-enhanced section **118**G may not include portions of the other two (2) of the pillar structures **122** of the pillar group **124**. In some embodiments, the other two (2) of the pillar structures **122** include the first pillar structure **122**A and the second pillar structure **122**B. The seventh V.sub.t-enhanced section **118**G may be substantially free of conductivity-enhancing species (e.g., P-type dopants, N-type dopants) therein. Select transistors **123** within the seventh V.sub.t-enhanced section **118**G may be configured (e.g., programmed) to have substantially the same V.sub.t characteristics as one another.

(54) Within the horizontal area of the sub-section B.sub.1 of the microelectronic device structure **100**, one eighth V.sub.t-enhanced section **118**H may encompass portions of the second sub-block

114B and the third sub-block **114**C of the third sub-block group **116**C. The eighth V.sub.t-enhanced section **118**H may not encompass portions of either of the first sub-block **114**A and the fourth sub-block **114**D of the third sub-block group **116**C. The eighth V.sub.t-enhanced section **118**H may comprise portions of the second SGP bar tier **110**J doped with at least one conductivity-enhancing species, as well as portions of two (2) of the pillar structures **122** of the pillar group **124** at the vertical position of the second SGP bar tier **110**J. In some embodiments, the two (2) of the pillar structures **122** include the second pillar structure **122**B and the third pillar structure **122**C. The eighth V.sub.t-enhanced section **118**H may not include portions of the other two (2) of the pillar structures **122** of the pillar group **124**. In some embodiments, the other two (2) of the pillar structures **122** include the first pillar structure **122**A and the fourth pillar structure **122**D. The eighth V.sub.t-enhanced section **118**H may be substantially free of conductivity-enhancing species (e.g., P-type dopants, N-type dopants) therein. Select transistors **123** within the eighth V.sub.t-enhanced section **118**H may be configured (e.g., programmed) to have substantially the same V.sub.t characteristics as one another.

(55) Within the horizontal area of the sub-section B.sub.1 of the microelectronic device structure **100**, one ninth V.sub.t-enhanced section **118**I may encompass portions of the first sub-block **114**A and the second sub-block **114**B of the third sub-block group **116**C. The ninth V.sub.t-enhanced section **118**I may not encompass portions of either of the third sub-block **114**C and the fourth sub-block **114**D of the third sub-block group **116**C. The ninth V.sub.t-enhanced section **118**I may comprise portions of the first SGP tier **110**K doped with at least one conductivity-enhancing species, as well as portions of two (2) of the pillar structures **122** of the pillar group **124** at the vertical position of the first SGP tier **110**K. In some embodiments, the two (2) of the pillar structures **122** include the first pillar structure **122**A and the second pillar structure **122**B. The ninth V.sub.t-enhanced section **118**I may not include portions of the other two (2) of the pillar structures **122** of the pillar group **124**. In some embodiments, the other two (2) of the pillar structures **122** include the third pillar structure **122**C and the fourth pillar structure **122**D. The ninth V.sub.t-enhanced section **118**I may be doped with one or more conductivity-enhancing species (e.g., at least one P-type dopant, such as one or more of boron (B), aluminum (Al), and gallium (Ga)). In some embodiments, the ninth V.sub.t-enhanced section **118**I is doped with B. The ninth V.sub.t-enhanced section **118**I may permit select transistors **123** associated with the two (2) of the pillar structures **122** at the vertical elevation of the first SGP tier **110**K to have substantially the same V.sub.t characteristics as one another; and may also permit select transistors **123** associated with the other two (2) of the pillar structures **122** to have substantially the same V.sub.t characteristics as one another that are different than the V.sub.t characteristics of the two (2) of the pillar structures **122**. In some embodiments, the select transistors **123** associated with the two (2) of the pillar structures **122** at the vertical elevation of the first SGP tier **110**K comprise relatively higher V.sub.t transistors; and the select transistors **123** associated with the other two (2) of the pillar structures **122** at the vertical elevation of the first SGP tier **110**K comprise relatively lower V.sub.t transistors.

(56) Within the horizontal area of the sub-section B.sub.1 of the microelectronic device structure **100**, one tenth V.sub.t-enhanced section **118**J may encompass a portion of the first sub-block **114**A and another tenth V.sub.t-enhanced section **118**J may encompass a portion of the fourth sub-block **114**D. Neither of the tenth V.sub.t-enhanced sections **118**J may encompass portions of either of the second sub-block **114**B and the third sub-block **114**C of the third sub-block group **116**C. The tenth V.sub.t-enhanced sections **118**J may comprise portions of the second SGP tier **110**L doped with at least one conductivity-enhancing species, as well as portions of two (2) of the pillar structures **122** of the pillar group **124** at the vertical position of the second SGP tier **110**L. In some embodiments, the two (2) of the pillar structures **122** include the first pillar structure **122**A and the fourth pillar structure **122**D. The tenth V.sub.t-enhanced sections **118**J may not include portions of the other two (2) of the pillar structures **122** of the pillar group **124**. In some embodiments, the other two (2) of the pillar structures **122** include the second pillar structure **122**B and the third pillar structure **122**C.

The tenth V.sub.t-enhanced sections **118**J may be doped with one or more conductivity-enhancing species (e.g., at least one P-type dopant, such as one or more of B, Al, and Ga). In some embodiments, the tenth V.sub.t-enhanced sections **118**J are doped with B. The tenth V.sub.t-enhanced sections **118**J may permit select transistors **123** associated with the two (2) of the pillar structures **122** at the vertical elevation of the second SGP tier **110**L to have substantially the same V.sub.t characteristics as one another; and may also permit select transistors **123** associated with the other two (2) of the pillar structures **122** to have substantially the same V.sub.t characteristics as one another that are different than the V.sub.t characteristics of the two (2) of the pillar structures **122**. In some embodiments, the select transistors **123** associated with the two (2) of the pillar structures **122** at the vertical elevation of the second SGP tier **110**L comprise relatively higher V.sub.t transistors; and the select transistors **123** associated with the other two (2) of the pillar structures **122** at the vertical elevation of the second SGP tier **110**L comprise relatively lower V.sub.t transistors.
(57) Collectively referring to FIGS. **1**A and **1**B, the SGD plug tier **127** may include multiple (e.g., more than one, a plurality of) SGD plug structures **126**. The SGD plug structures **126** may vertically overlie the pillar structures **122**, and may individually horizontally extend across and between four (4) of the pillar structures **122** of an individual pillar group **124**. An individual SGD plug structure **126** may be in electrical communication with each of the four (4) of the pillar structures **122** of an individual pillar group **124**. For example, as shown in FIG. **1**A, within the horizontal area of the sub-section B.sub.1 of the microelectronic device structure **100**, one (1) of the SGD plug structures **126** may vertically overlie and be coupled to each of the first pillar structure **122**A, the second pillar structure **122**B, the third pillar structure **122**C, and the fourth pillar structure **122**D of an individual pillar group **124** within the third sub-block group **116**C of an individual block **164** (FIG. **1**B) of the stack structure **104**. The first pillar structure **122**A may be located within the first sub-block **114**A of the third sub-block group **116**C; the second pillar structure **122**B may be located within the second sub-block **114**B of the third sub-block group **116**C; the third pillar structure **122**C may be located within the third sub-block **114**C of the third sub-block group **116**C; and the fourth pillar structure **122**D may be located within the fourth sub-block **114**D of the third sub-block group **116**C. Accordingly, an individual SGD plug structure **126** may be coupled to each of four (4) pillar structures **122** (and, hence, the select transistors **123** and the strings of memory cells **121** that are operatively associated with each of the four (4) of the pillar structures **122**) in different sub-blocks **114** than one another within an individual block **164** (FIG. **1**B).
(58) Referring to FIG. **1**B, some of the SGD plug structures **126** may horizontally extend in parallel with one another, and some other of the SGD plug structures **126** may horizontally extend in series with one another. SGD plug structures **126** within a horizontal area of the same sub-block group **116** (e.g., the first sub-block group **116**A, the second sub-block group **116**B, the third sub-block group **116**C, or the fourth sub-block group **116**D) as one another may horizontally extend in parallel with one another in a direction acutely angled relative to each of the X-direction and the Y-direction shown in FIG. **1**B. In addition, at least some SGD plug structures **126** within horizontal areas of different sub-block groups **116** than one another may horizontally extend in series with one another in the direction acutely angled relative to each of the X-direction and the Y-direction shown in FIG. **1**B.
(59) The SGD plug structures **126** may individually be formed of and include conductive material. By way of non-limiting example, the SGD plug structures **126** may each individually formed of and include semiconductor material (e.g., polycrystalline silicon) doped with doped with at least one conductivity-enhancing dopant (e.g., at least one N-type dopant, such as one or more of P, As, Sb, and Bi; at least one P-type dopant, such as one or more of B, Al, and Ga). In some embodiments, the SGD plug structures **126** may each individually formed of and include N-type polycrystalline silicon (e.g., polycrystalline silicon doped with at least one N-type dopant, such as

polycrystalline silicon doped with P). Each of the SGD plug structures **126** may individually be substantially homogeneous, or one or more of the SGD plug structures **126** may individually be substantially heterogeneous. In some embodiments, each of the SGD plug structures **126** is substantially homogeneous.

(60) Referring to FIG. **1**A, the SGD plug structures **126**, portions of the sense node tier section **111**C within horizontal areas of the SGD plug structures **126**, and portions of the pillar structures **122** within horizontal areas of the SGD plug structures **126** and within vertical boundaries of the sense node tier section **111**C may together form sense nodes **112** of the microelectronic device structure **100**. Each sense node **112** may individually be positioned within a horizontal area of an individual sub-block group **116** of an individual block **164** (FIG. **1**B) of the stack structure **104**, and may vertically overlie the stacked SGD tier section **111**B (and, hence, the access line tier section **111**A, and the strings of memory cells **121**) of the stack structure **104**. An individual sense node **112** may include a group of the select transistors **123** within a horizontal area of the SGD plug structure **126** of sense node **112** and within vertical boundaries of the sense node tier section **111**C. For example, an individual sense node **112** within the horizontal area of the sub-section B.sub.1 of the microelectronic device structure **100** shown in FIG. **1**A may include one (1) of the SGD plug structures **126**, as well as the select transistors **123** formed at the intersections of the four (4) pillar structures **122** (e.g., the first pillar structure **122**A, the second pillar structure **122**B, the third pillar structure **122**C, and the fourth pillar structure **122**D) coupled to the SGD plug structures **126** and the conductive structures **106** of tiers **110** (e.g., the first SGP bar tier **110**I, the second SGP bar tier **110**J, the first SGP tier **110**K, the second SGP tier **110**L) of the sense node tier section **111**C and the four (4) pillar structures **122** (e.g., the first pillar structure **122**A, the second pillar structure **122**B, the third pillar structure **122**C, and the fourth pillar structure **122**D). The sense nodes **112** within horizontal areas of blocks **164** (FIG. **1**B) of the stack structure **104** may facilitate capacitive-sense on block functionality for the microelectronic device structure **100**, as described in further detail below.

(61) Still referring to FIG. **1**A, the microelectronic device structure **100** further includes a channel tier **132** including channel structures **134** vertically interposed between the SGD plug tier **127** and the selector tiers **138**. The channel structures **134** may horizontally overlap the SGD plug structures **126**. In some embodiments, the channel structures **134** are substantially confined within horizontal areas of the SGD plug structures **126**, and exhibit horizontal areas relatively smaller than the horizontal areas of the SGD plug structures **126**. For example, an individual channel structure **134** may horizontally extend across a portion of an individual SGD plug structure **126** that covers less than all of the pillar structures **122** coupled to the SGD plug structure **126**, such as only three (3) of the four (4) pillar structures **122** coupled to the SGD plug structure **126**. In some embodiments, an individual channel structure **134** horizontally overlaps three (3) of the pillar structures **122** (e.g., the second pillar structure **122**B, the third pillar structure **122**C, the fourth pillar structure **122**D) coupled to an individual SGD plug structure **126**, but does not substantially horizontally overlap a final one (1) of the pillar structures **122** (e.g., the first pillar structure **122**A) coupled to the SGD plug structure **126**.

(62) Referring to FIG. **1**B, some of the channel structures **134** may horizontally extend in parallel with one another, and some other of the channel structures **134** may horizontally extend in series with one another. Channel structures **134** within a horizontal area of the same sub-block group **116** (e.g., the first sub-block group **116**A, the second sub-block group **116**B, the third sub-block group **116**C, or the fourth sub-block group **116**D) as one another may horizontally extend in parallel with one another in a direction acutely angled relative to each of the X-direction and the Y-direction shown in FIG. **1**B. In addition, at least some channel structures **134** within horizontal areas of different sub-block groups **116** than one another may horizontally extend in series with one another in the direction acutely angled relative to each of the X-direction and the Y-direction shown in FIG. **1**B.

(63) The channel structures **134** of the channel tier **132** may individually be formed of and include at least one semiconductor material, such as one or more of silicon, germanium, at least one compound semiconductor material, and at least one oxide semiconductor material. In some embodiments, the channel structures **134** are formed of and include polycrystalline silicon. In additional embodiments, the channel structures **134** are formed of and include oxide semiconductor material. Each of the channel structures **134** may individually be substantially homogeneous, or one or more of the channel structures **134** may individually be substantially heterogeneous. In some embodiments, each of the channel structures **134** is substantially homogeneous. The semiconductor material of the channel structures **134** may be doped with one or more conductivity-enhancing species (e.g., P-type dopant, N-type dopant), or the semiconductor material of the channel structures **134** may not be doped one or more conductivity-enhancing species. In some embodiments, the channel structures **134** are each individually formed of and include P-type polycrystalline silicon (e.g., polycrystalline silicon doped with at least one P-type dopant, such as polycrystalline silicon doped with B).
(64) Still referring to FIG. **1**A, a gate dielectric tier **129** including gate dielectric structures **130** vertically interposed between the channel tier **132** and the SGD plug tier **127**. The gate dielectric structures **130** may horizontally extend across at least the channel structures **134** of the channel tier **132** and portions of the SGD plug structures **126** of the SGD plug tier **127**, and may vertically extend from and between the channel structures **134** and the SGD plug structures **126**. The gate dielectric structures **130** may be formed of and include dielectric material, such as one or more of at least one dielectric oxide material (e.g., one or more of SiO.sub.x, AlO.sub.x, phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass), at least one dielectric nitride material (e.g., SiN.sub.y), and at least one low-K dielectric material (e.g., one or more of silicon oxycarbide (SiO.sub.xC.sub.y), silicon oxynitride (SiO.sub.xN.sub.y), hydrogenated silicon oxycarbide (SiC.sub.xO.sub.yH.sub.z), and silicon oxycarbonitride (SiO.sub.xC.sub.zN.sub.y)). The gate dielectric structures **130** may be substantially homogeneous, or the gate dielectric structures **130** may be heterogeneous. In some embodiments, the gate dielectric structure **130** is formed of and includes SiO.sub.x (e.g., SiO.sub.2).
(65) The SGD plug structures **126**, the gate dielectric structures **130**, and the channel structures **134** may together form horizontal sense transistors **136** (e.g., planar sense transistors, laterally oriented sense transistors) of the microelectronic device structure **100**. Each horizontal sense transistor **136** may be a thin film transistor (TFT). The SGD plug structures **126** may serve as gate electrodes for the horizontal sense transistors **136**. Each horizontal sense transistor **136** may individually include a SGD plug structure **126**, a channel structure **134** horizontally overlapping the SGD plug structure **126**, and gate dielectric structures **130** vertically interposed between the SGD plug structure **126** and the channel structure **134**.
(66) Still referring to FIG. **1**A, the selector tiers **138** may vertically overlie the channel tier **132** (and, hence, the horizontal sense transistors **136**). The selector tiers **138** may include a first selector tier **138**A, a second selector tier **138**B vertically overlying the first selector tier **138**A, and a third selector tier **138**C vertically overlying the second selector tier **138**B. Each of the selector tiers **138** may individually include an additional conductive structure **140** (e.g., gate electrode). Insulative material (e.g., dielectric oxide material, such as SiO.sub.x) may be vertically interleaved with the additional conductive structures **140** of the selector tiers **138**. The additional conductive structures **140** of the selector tiers **138** may be confined within horizontal areas of the blocks **164** (FIG. **1**B) of the stack structure **104**. In some embodiments, the additional conductive structures **140** of the selector tiers **138** are formed substantially simultaneously with the formation of the conductive structures **106** of the tiers **110** of the stack structure **104** (e.g., during so-called “replacement gate” or “gate last” processes), as described in further detail below.
(67) The additional conductive structures **140** of the selector tiers **138** may individually be formed of and include conductive material. By way of non-limiting example, the additional conductive

structures **140** may individually be formed of and include one or more of at least one metal, at least one alloy, and at least one conductive metal-containing material (e.g., a conductive metal nitride, a conductive metal silicide, a conductive metal carbide, a conductive metal oxide). In some embodiments, the additional conductive structures **140** are individually formed of and include one or more of W, Ru, Mo, and TiN.sub.y. Each of the additional conductive structures **140** may individually be substantially homogeneous, or one or more of the additional conductive structures **140** may individually be substantially heterogeneous. In some embodiments, each of the additional conductive structures **140** is substantially homogeneous.

(68) Within a horizontal area of an individual block **164** (FIG. **1**B), one (1) of the selector tiers **138** (e.g., the first selector tier **138**A) may be employed as a write selector tier, another one (1) of the selector tiers **138** (e.g., the second selector tier **138**B) may be employed as a read selector tier, and an additional one (1) of the selector tiers **138** (e.g., the third selector tier **138**C) may be employed as a block selector tier. In some embodiments, the first selector tier **138**A is employed as a write selector tier, and an individual additional conductive structure **140** thereof is employed as write select gate electrode; the second selector tier **138**B is employed as a read selector tier, and an individual additional conductive structure **140** thereof is employed as read select gate electrode; and the third selector tier **138**C is employed as a block selector tier, and an individual additional conductive structure **140** thereof is employed as block select gate electrode.

(69) With continued reference to FIG. **1**A, the microelectronic device structure **100** further includes write select pillar structures **144** vertically extending through the selector tiers **138** and to or into the SGD plug structures **126**. The write select pillar structures **144** may be positioned with horizontal areas of the SGD plug structures **126**, and may be horizontally offset from the channel structures **134**. Each of the SGD plug structures **126** may individually include one (1) of the write select pillar structures **144** in contact (e.g., physical contact, electrical contact) therewith. As shown in FIG. **1**A, in some embodiments, an individual write select pillar structure **144** horizontally overlaps (e.g., is substantially horizontally centered about) one (1) of the pillar structures **122** (e.g., the first pillar structure **122**A) coupled to the SGD plug structure **126** in contact with the write select pillar structure **144**.

(70) Still referring to FIG. **1**A, the microelectronic device structure **100** further includes first read select pillar structures **146** (e.g., source-side read select pillars) and second read select pillar structures **148** (e.g., drain-side read select pillars) vertically extending through the selector tiers **138** and to or into the channel structures **134**. The first read select pillar structures **146** and the second read select pillar structures **148** may be positioned with horizontal areas of the channel structures **134**, and may be horizontally offset from one another and the channel structures **134**. Each of the channel structures **134** may individually include one (1) of the first read select pillar structures **146** and one (1) of the second read select pillar structures **148** in contact (e.g., physical contact, electrical contact) therewith. Accordingly, each of the horizontal sense transistors **136** may individually include one (1) of the first read select pillar structures **146** and one (1) of the second read select pillar structures **148** in electrical communication therewith. In some embodiments, for an individual horizontal sense transistor **136**, a first read select pillar structure **146** is employed a source-side read select pillar that contacts a source region of the channel structure **134**; and a second read select pillar structure **148** is employed a drain-side read select pillar that contacts a drain region of the channel structure **134**. Within a horizontal area of an individual SGD plug structure **126**, a horizontal distance between the second read select pillar structure **148** and the write select pillar structure **144** may be less than a horizontal distance between the first read select pillar structure **146** and the write select pillar structure **144**. As shown in FIG. **1**A, in some embodiments, an individual first read select pillar structure **146** horizontally overlaps (e.g., is substantially horizontally centered about) one (1) of the pillar structures **122** (e.g., the fourth pillar structure **122**D) coupled to the SGD plug structure **126** horizontally overlapping the first read select pillar structure **146**; and an individual second read select pillar structure **148** horizontally overlaps (e.g.,

is substantially horizontally centered about) an additional one (1) of the pillar structures **122** (e.g., the second pillar structure **122**B) coupled to the SGD plug structure **126** horizontally overlapping the second read select pillar structure **148**.

(71) The write select pillar structures **144**, the first read select pillar structures **146**, and the second read select pillar structures **148** may each individually be formed of and include a stack of materials. By way of non-limiting example, the write select pillar structures **144**, the first read select pillar structures **146**, and the second read select pillar structures **148** may each individually be formed of and include a gate dielectric material (e.g., SiO.sub.x, such as SiO.sub.2), a channel material (e.g., silicon, such as polycrystalline silicon), and a dielectric fill (e.g., SiO.sub.x, such as SiO.sub.2; SiN.sub.y, such as Si.sub.3N.sub.4; air). The gate dielectric material may be formed on or over surfaces of the additional conductive structures **140** and the insulative material of the selector tiers **138** at least partially defining horizontal boundaries of the write select pillar structures **144**, the first read select pillar structures **146**, and the second read select pillar structures **148**; the channel material may be horizontally surrounded by the gate dielectric material; and the dielectric fill material may be horizontally surrounded by the channel material.

(72) Still referring to FIG. **1**A, intersections of the write select pillar structures **144** and the additional conductive structures **140** of individual selector tiers **138** may define write selector transistors **131** of the microelectronic device structure **100**. By way of non-limiting example, intersections of the write select pillar structures **144** and the additional conductive structure **140** (e.g., serving as a write select gate) of first selector tier **138**A the may define write selector transistors **131** at the vertical elevation of the first selector tier **138**A. An individual write selector transistor **131** may comprise a vertical transistor (e.g., a vertically oriented transistor) including a channel region vertically offset from source/drain regions. Channel regions of the write selector transistors **131** may be positioned within vertical boundaries of the additional conductive structure **140** of an individual selector tier **138**; and source/drain regions of the write selector transistors **131** may vertically neighbor the channel regions, and may be vertically offset from the additional conductive structure **140**. In some embodiments, the write selector transistors **131** comprise metal-oxide-semiconductor (MOS) transistors. Write selector transistors **131** within vertical boundaries of selector tiers **138** employed as write selector tiers (e.g., the first selector tier **138**A) may have different V.sub.t characteristics than write selector transistors **131** of selector tiers **138** employed as read selector tiers (e.g., the second selector tier **138**B). For example, write selector transistors **131** within vertical boundaries of the first selector tier **138**A may comprise relatively higher V.sub.t transistors, and write selector transistors **131** within vertical boundaries of the second selector tier **138**B may comprise relatively lower V.sub.t transistors.

(73) Intersections of the additional conductive structures **140** of individual selector tiers **138** and each of the first read select pillar structures **146** and the second read select pillar structures **148** may define read selector transistors **135** of the microelectronic device structure **100**. By way of non-limiting example, intersections of the additional conductive structure **140** of the second selector tier **138**B and each of the first read select pillar structures **146** and the second read select pillar structures **148** may define read selector transistors **135** at the vertical elevation of the second selector tier **138**B. An individual read selector transistor **135** may comprise a vertical transistor (e.g., a vertically oriented transistor) including a channel region vertically offset from source/drain regions. Channel regions of the read selector transistors **135** may be positioned within vertical boundaries of the additional conductive structure **140** of an individual selector tier **138**; and source/drain regions of the read selector transistors **135** may vertically neighbor the channel regions, and may be vertically offset from the additional conductive structure **140**. In some embodiments, the read selector transistors **135** comprise MOS transistors. Read selector transistors **135** within vertical boundaries of selector tiers **138** employed as read selector tiers (e.g., the second selector tier **138**B) may have different V.sub.t characteristics than read selector transistors **135** of selector tiers **138** employed as write selector tiers (e.g., the first selector tier **138**A). For example,

read selector transistors **135** within vertical boundaries of the second selector tier **138**B may comprise relatively higher V.sub.t transistors, and read selector transistors **135** within vertical boundaries of the first selector tier **138**A may comprise relatively lower V.sub.t transistors.

(74) Write selector transistors **131** and read selector transistors **135** within vertical boundaries of an individual selector tier **138** employed as a block selector tier may be employed for block select operations for a microelectronic device including the microelectronic device structure **100**. By way of non-limiting example, if the third selector tier **138**C is employed as a block selector tier of the microelectronic device structure **100**, the write selector transistors **131** and read selector transistors **135** within vertical boundaries of the third selector tier **138**C may be employed for block select operations for a microelectronic device including the microelectronic device structure **100**. Within a horizontal area of an individual block **164** (FIG. **1**B), write selector transistors **131** and read selector transistors **135** within vertical boundaries of an individual selector tier **138** employed as a block selector tier (e.g., the third selector tier **138**C) may have substantially the same V.sub.t characteristics as one another. For example, within a horizontal area of an individual block **164** (FIG. **1**B), write selector transistors **131** and read selector transistors **135** within vertical boundaries of the third selector tier **138**C may all comprise relatively higher V.sub.t transistors.

(75) Still referring to FIG. **1**A, different portions of the selector tiers **138** (e.g., the first selector tier **138**A, the second selector tier **138**B, the third selector tier **138**C), the write select pillar structures **144**, the first read select pillar structures **146**, and the second read select pillar structures **148** may be doped with one or more conductivity-enhancing species (e.g., one or more P-type dopants, such as one or more of B, Al, and Ga; one or more N-type dopants, such as one or more of P, As, Sb, and Bi). By way of non-limiting example, within horizontal areas of the blocks **164** (FIG. **1**B), the microelectronic device structure **100** may include doped sections **142** at vertical positions of at least some of the selector tiers **138**. Each doped section **142** may individually include a portion of a selector tier **138** doped with at least one conductivity-enhancing species (e.g., at least one P-type dopant, at least one N-type dopant), as well as portions of one or more a write select pillar structure **144**, a first read select pillar structure **146**, and a second read select pillar structure **148** doped with the at least one conductivity-enhancing species. An individual doped section **142** may be substantially confined within horizontal boundaries of an individual sub-block group **116** of the block **164** (FIG. **1**B), or may horizontally extend at least partially across and between at least two (2) horizontally neighboring sub-block groups **116** of the block **164** (FIG. **1**B). The doped sections **142** of the microelectronic device structure **100** may be used to provide different select transistors (e.g., write selector transistors **131**, read selector transistors **135**, block select transistors) within the selector tiers **138** with desired V.sub.t characteristics, as described in further detail below. For example, as a result of the doped sections **142**, some of the write selector transistors **131** and read selector transistors **135** within boundaries (e.g., vertical boundaries, horizontal boundaries) of doped sections **142** may be relatively higher V.sub.t transistors, while some other of the write selector transistors **131** and read selector transistors **135** outside of the boundaries of doped sections **142** may be relatively lower V.sub.t transistors.

(76) Within an individual block **164** (FIG. **1**B) of the stack structure **104**, the doped sections **142** may be configured such that the write selector transistors **131**, the read selector transistors **135**, and the block select transistors have desired V.sub.t characteristics for use and operation of a microelectronic device including the microelectronic device structure **100**. For example, the doped sections **142** may be configured such that read selector transistors **135** substantially vertically aligned with one another (e.g., within the same selector tier **138** as one another) have substantially the same V.sub.t characteristics (e.g., are set to the same V.sub.t levels) as one another, and different V.sub.t characteristics (e.g., are set to different V.sub.t levels) than write selector transistors **131** substantially vertically aligned therewith (e.g., within the same selector tier **138** as one another). As another example, the doped sections **142** may be configured such that some vertically neighboring transistors operatively associated with the same write select pillar structure **144**, the same first read

select pillar structure **146**, or the same second read select pillar as one another have different V.sub.t characteristics (e.g., different V.sub.t levels) than one another.

(77) As shown in FIG. **1**A, in some embodiments, the doped sections **142** of the microelectronic device structure **100** include first doped sections **142**A at a vertical elevation of the first selector tier **138**A; second doped sections **142**B at a vertical elevation of the second selector tier **138**B; and third doped sections **142**C at a vertical elevation of the third selector tier **138**C. Within a horizontal area of an individual SGD plug structure **126**, different doped sections **142** may have horizontal positions than one another and/or may have different horizontal dimensions than one another. Non-limiting examples of configurations of some of the doped sections **142** within a horizontal area of the sub-section B.sub.1 of the microelectronic device structure **100** shown in FIGS. **1**A and **1**B are described in further detail below.

(78) Referring to FIG. **1**A, within a horizontal area of the sub-section B.sub.1 of the microelectronic device structure **100**, a first doped section **142**A may comprise a portion of at least the additional conductive structure **140** of the first selector tier **138**A doped with at least one conductivity-enhancing species, as well as a portion of the write select pillar structure **144** at the vertical position of the additional conductive structure **140** of the first selector tier **138**A. The first doped section **142**A may not include portions of the first read select pillar structure **146** and the second read select pillar structure **148** at the vertical position of the additional conductive structure **140** of the first selector tier **138**A. For example, the first doped section **142**A may not substantially horizontally overlap the channel structure **134** in contact with the first read select pillar structure **146** and the second read select pillar structure **148**. The first doped section **142**A may permit a write selector transistor **131** at the vertical position of the first selector tier **138**A to have different V.sub.t characteristics than additional transistors (e.g., read selector transistors **135**) at the vertical position of the first selector tier **138**A. In some embodiments, the conductivity-enhancing species of the first doped section **142**A includes P-type dopant (e.g., B).

(79) Within a horizontal area of the sub-section B.sub.1 of the microelectronic device structure **100**, a second doped section **142**B may comprise portions of at least the additional conductive structure **140** of the second selector tier **138**B doped with at least one conductivity-enhancing species, as well as a portion of the first read select pillar structure **146** and the second read select pillar structure **148** at the vertical position of the additional conductive structure **140** of the second selector tier **138**B. The second doped section **142**B may not include a portion of the write select pillar structure **144** at the vertical position of the additional conductive structure **140** of the second selector tier **138**B. For example, the second doped section **142**B may substantially horizontally confined within the horizontal area of the channel structure **134** in contact with the first read select pillar structure **146** and the second read select pillar structure **148**. The second doped section **142**B may permit read selector transistors **135** at the vertical position of the second selector tier **138**B to have substantially the same V.sub.t characteristics as one another and different V.sub.t characteristics than additional transistors (e.g., a write selector transistor **131**) at the vertical position of the second selector tier **138**B. In some embodiments, the conductivity-enhancing species of the second doped section **142**B includes P-type dopant (e.g., B). Optionally, an additional portion of at least the additional conductive structure **140** of the second selector tier **138**B and a portion of the write select pillar structure **144** may be doped with an additional conductivity-enhancing species (e.g., N-type dopant, such as P) having a conductivity type different than the conductivity-enhancing species (e.g., P-type dopant, such as B) included in the second doped section **142**B.

(80) Within a horizontal area of the sub-section B.sub.1 of the microelectronic device structure **100**, a third doped section **142**C may comprise portions of at least the additional conductive structure **140** of the third selector tier **138**C doped with at least one conductivity-enhancing species, as well as a portion of the write select pillar structure **144**, the first read select pillar structure **146**, and the second read select pillar structure **148** at the vertical position of the additional conductive structure

140 of the third selector tier **138**C. For example, the third doped section **142**C may continuously horizontally extend across and beyond the horizontal area of the SGD plug structure **126** within the sub-section B.sub.1 of the microelectronic device structure **100**. The third doped section **142**C may permit additional transistors (e.g., write selector transistors **131**, read selector transistors **135**, block select transistors) at the vertical position of the third selector tier **138**C to have substantially the same V.sub.t characteristics as one another. In some embodiments, the conductivity-enhancing species of the third doped section **142**C includes P-type dopant (e.g., B).

(81) Referring collectively to FIGS. **1**A and **1**B, the conductive routing tier **150** (FIG. **1**A) includes read source line structures **152** and local strap structures **154** vertically overlying the selector tiers **138**. The read source line structures **152** may individually vertically overlie, horizontally overlap, and contact (e.g., physically contact, electrically contact) multiple of the first read select pillar structures **146** (e.g., source-side read select pillar structures) a horizontal area of and individual block **164** (FIG. **1**B) of the stack structure **104**. The local strap structures **154** may individually vertically overlie, horizontally extend from and between, and contact (e.g., physically contact, electrically contact) a write select pillar structure **144** and a second read select pillar structure **148** within a horizontal area of an individual SGD plug structure **126**. The read source line structures **152** and local strap structures **154** may be substantially vertically aligned with one another within the conductive routing tier **150**.

(82) As shown in FIG. **1**B, the read source line structures **152** may horizontally extend in parallel within one another and rows of the first read select pillar structures **146** (as well as rows of the second read select pillar structures **148**, rows of the write select pillar structures **144**, and rows of the pillar structures **122**) in the X-direction. Each of the read source line structures **152** may individually be coupled to the first read select pillar structures **146** of at least one (1) row of the first read select pillar structures **146** extending in the X-direction. As depicted in FIG. **1**B, in some embodiments, each source line structure **152** is individually coupled to two (2) rows of the first read select pillar structures **146**. A first of the two (2) rows of the first read select pillar structures **146** may be positioned within a horizontal area of one (1) of the sub-block groups **116** of an individual block **164**; and a second of the two (2) rows of the first read select pillar structures **146** may be positioned within a horizontal area of another one (1) of the sub-block groups **116** of block **164** horizontally neighboring the one (1) of the sub-block groups **116**.

(83) Referring to FIGS. **1**A and **1**B, the local strap structures **154** may individually couple a write select pillar structure **144** within a horizontal area of an individual SGD plug structure **126** to a second read select pillar structure **148** within the horizontal area of SGD plug structure **126**. As shown FIG. **1**B, some of the local strap structures **154** may horizontally extend in parallel with one another (and some of the SGD plug structures **126** and some of the channel structures **134**), and some other of the local strap structures **154** may horizontally extend in series with one another (and some other of the SGD plug structures **126** and some other of the channel structures **134**). Local strap structures **154** within a horizontal area of the same sub-block group **116** (e.g., the first sub-block group **116**A, the second sub-block group **116**B, the third sub-block group **116**C, or the fourth sub-block group **116**D) as one another may horizontally extend in parallel with one another in a direction acutely angled relative to each of the X-direction and the Y-direction shown in FIG. **1**B. In addition, at least some local strap structures **154** within horizontal areas of different sub-block groups **116** than one another may horizontally extend in series with one another in the direction acutely angled relative to each of the X-direction and the Y-direction shown in FIG. **1**B.

(84) The read source line structures **152** and the local strap structures **154** may each individually be formed of and include conductive material. By way of non-limiting example, the read source line structures **152** and the local strap structures **154** may each individually be formed of and include one or more of at least one metal, at least one alloy, and at least one conductive metal-containing material (e.g., a conductive metal nitride, a conductive metal silicide, a conductive metal carbide, a conductive metal oxide). In some embodiments, the read source line structures **152** and the local

strap structures **154** are each individually formed of and include one or more of W, Ru, Mo, and TiN.sub.y. Each of the read source line structures **152** and the local strap structures **154** may individually be substantially homogeneous, or one or more of the read source line structures **152** and/or and one or more of the local strap structures **154** may individually be substantially heterogeneous. In some embodiments, each of the read source line structures **152** and each of the local strap structures **154** is substantially homogeneous.
(85) Referring to FIG. **1**B, in some embodiments, within boundaries of an individual sub-block group **116**, all of the channel structures **134** are substantially aligned with one another in the Y-direction, all of the write select pillar structures **144** are substantially aligned with one another in the Y-direction, all of the first read select pillar structures **146** are substantially aligned with one another in the Y-direction, all of the second read select pillar structures **148** are substantially aligned with one another in the Y-direction, and all of the local strap structures **154** are substantially aligned with one another in the Y-direction. In additional embodiments, within boundaries of an individual sub-block group **116**, some of the channel structures **134** are horizontally offset from one another in the Y-direction, some of the write select pillar structures **144** are horizontally offset from one another in the Y-direction, some of the first read select pillar structures **146** are horizontally offset from one another in the Y-direction, some of the second read select pillar structures **148** are horizontally offset from one another in the Y-direction, and/or some of the local strap structure **154** are horizontally offset from one another in the Y-direction. For example, horizontal positions in the Y-direction of channel structures **134**, write select pillar structures **144**, first read select pillar structures **146**, second read select pillar structures **148**, and local strap structure **154** operatively associated with SGD plug structures **126** (and, hence, horizontal sense transistors **136**) horizontally neighboring one another in the X-direction within boundaries of an individual sub-block group **116** may be horizontally inverted in the Y-direction relative to another. By way of non-limiting example, within boundaries of an individual sub-block group **116**, a write select pillar structure **144** operatively associated with a first SGD plug structure **126** of two (2) horizontally neighboring SGD plug structures **126** may be substantially horizontally aligned in the Y-direction with a first read select pillar structure **146** operatively associated with a second SGD plug structure **126** of the two (2) horizontally neighboring SGD plug structures **126**; a write select pillar structure **144** operatively associated with the second SGD plug structure **126** may be substantially horizontally aligned in the Y-direction with a first read select pillar structure **146** operatively associated with the first SGD plug structure **126**; a second read select pillar structure **148** operatively associated with the first SGD plug structure **126** may be completely horizontally offset in the Y-direction from a second read select pillar structure **148** operatively associated with the second SGD plug structure **126**; a channel structure **134** operatively associated with the first SGD plug structure **126** may be partially horizontally offset in the Y-direction from a channel structure **134** operatively associated with the second SGD plug structure **126**; and a local strap structure **154** operatively associated with the first SGD plug structure **126** may be completely horizontally offset in the Y-direction from a local strap structure **154** operatively associated with the second SGD plug structure **126**.
(86) Referring collectively to FIGS. **1**A and **1**B, the digit line tier **156** (FIG. **1**A) includes digit line structures **158** vertically overlying the conductive routing tier **150**. As shown in FIG. **1**B, the digit line structures **158** may horizontally extend in parallel with one another in the Y-direction. The digit line structures **158** may horizontally extend perpendicular to the read source line structures **152**. The digit line structures **158** may be coupled to logic circuitry (e.g., page buffer circuitry) of a microelectronic device including the microelectronic device structure **100**. The logic circuitry may, for example, be included within the base structure **102** (FIG. **1**A) vertically underlying the stack structure **104**. In some embodiments, the digit line structures **158** are coupled to page buffer devices each individually including an arrangement of data cache circuitry (e.g., dynamic data cache (DDC) circuitry, primary data cache (PDC) circuitry, secondary data cache (SDC) circuitry, temporary data cache (TDC) circuitry), sense amplifier (SA) circuitry, and digit line pre-charge

circuitry. Optionally, isolation devices (e.g., isolation transistors) may be interposed between the digit line structures **158** and the page buffer devices at desirable locations along conductive paths extending from and between the digit line structures **158** and the page buffer devices. In some embodiments, the isolation devices comprise high-voltage-isolation (HVISO) transistors configured and operated to pass voltages greater than or equal to about 18V, such as within a range of from about 18V to about 25V. In additional embodiments, the isolation devices comprise low-voltage-isolation (LVISO) transistors configured and operated to substantially block applied voltages less than about 18 V while in an OFF state (e.g., an inactive state, a depletion state, a deselected state).

(87) The digit line structures **158** may each individually be formed of and include conductive material. By way of non-limiting example, the digit line structures **158** may each individually be formed of and include one or more of at least one metal, at least one alloy, and at least one conductive metal-containing material (e.g., a conductive metal nitride, a conductive metal silicide, a conductive metal carbide, a conductive metal oxide). In some embodiments, the digit line structures **158** are each individually formed of and include one or more of W, Ru, Mo, and TiN.sub.y. Each of the digit line structures **158** may individually be substantially homogeneous, or one or more of the digit line structures **158** may individually be substantially heterogeneous. In some embodiments, each of the digit line structures **158** is substantially homogeneous.

(88) Referring to FIG. **1**A, the microelectronic device structure **100** further includes digit line contact structures **160** vertically interposed between the digit line tier **156** and the conductive routing tier **150**. The digit line contact structures **160** may be configured to couple individual digit line structures **158** to individual local strap structures **154**. Each local strap structure **154** within a horizontal area of an individual block **164** (FIG. **1**B) of the stack structure **104** may individually be coupled to an individual digit line structure **158** by way of an individual digit line contact structure **160**. The digit line contact structure **160** may vertically extend from and between the local strap structure **154** and the digit line structure **158**.

(89) The digit line contact structures **160** may each individually be formed of and include conductive material. By way of non-limiting example, the digit line contact structures **160** may each individually be formed of and include one or more of at least one metal, at least one alloy, and at least one conductive metal-containing material (e.g., a conductive metal nitride, a conductive metal silicide, a conductive metal carbide, a conductive metal oxide). In some embodiments, the digit line contact structures **160** are each individually formed of and include one or more of W, Ru, Mo, and TiN.sub.y. Each of the digit line contact structures **160** may individually be substantially homogeneous, or one or more of the digit line contact structures **160** may individually be substantially heterogeneous.

(90) Still referring to FIG. **1**A, the microelectronic device structure **100** further includes at least one isolation material **162** covering and surrounding various features (e.g., materials, structures, devices) thereof. For example, the isolation material **162** may vertically overlie the stack structure **104**, and may at least partially cover, at least partially surround, and/or at least partially be interposed (e.g., horizontally interposed, vertically interposed) between additional features of the microelectronic device structure **100** vertically overlying the stack structure **104** (e.g., the SGD plug structures **126**, the channel structures **134**, the additional conductive structures **140**, the write select pillar structures **144**, the first read select pillar structures **146**, the second read select pillar structures **148**, the read source line structures **152**, the local strap structures **154**, the digit line contact structures **160**, the digit line structures **158**).

(91) The isolation material **162** may be formed of and include insulative material. By way of non-limiting example, the isolation material **162** may be formed of and include one or more of at least one dielectric oxide material (e.g., one or more of SiO.sub.x, phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass, AlO.sub.x, HfO.sub.x, NbO.sub.x, and TiO.sub.x), at least one dielectric nitride material (e.g., SiN.sub.y), at least one dielectric oxynitride

material (e.g., SiO.sub.xN.sub.y), at least one dielectric carboxynitride material (e.g., SiO.sub.xC.sub.zN.sub.y), and amorphous carbon. In some embodiments, the isolation material **162** is formed of and includes SiO.sub.x (e.g., SiO.sub.2). The isolation material **162** may be substantially homogeneous, or the isolation material **162** may be heterogeneous.

(92) During use and operation of a microelectronic device (e.g., a memory device, such as a 3D NAND Flash memory device) including the microelectronic device structure **100**, programming operations may be performed to set (e.g., program) V.sub.t levels of select transistors **123** within the SGD tier section **111**B and the sense node tier section **111**C of the stack structure **104**. Such operations are referred to herein as “SGPGM” operations. The SGPGM operations may be facilitated, in part, by the V.sub.t-enhanced sections **118** within the sense node tier section **111**C. The V.sub.t-enhanced sections **118** within the sense node tier section **111**C may, for example, be used to identify different pillar structures **122** within the horizontal area of an individual block **164** (FIG. **1**B) of the stack structure. As a non-limiting example, within a horizontal area of the sub-section B.sub.1 of the microelectronic device structure **100**, the tenth V.sub.t enhanced sections **118**J of the second SGP tier **110**L and the ninth V.sub.t-enhanced sections **118**I of the first SGP tier **110**K may be used to identify each of the first pillar structure **122**A, the second pillar structure **122**B, the third pillar structure **122**C, and the first pillar structure **122**A. The first pillar structure **122**A may be identified by the horizontal overlap of each of one (1) of the tenth V.sub.t-enhanced sections **118**J and one (1) of the ninth V.sub.t-enhanced sections **118**I with the first pillar structure **122**A, due to the impact thereof on the characteristics (e.g., channel types) of the select transistors **123** at the intersections of the first pillar structure **122**A and the first SGP tier **110**K and second SGP tier **110**L. The second pillar structure **122**B may be identified by the horizontal overlap of one (1) of the ninth V.sub.t-enhanced sections **118**I with the second pillar structure **122**B, along with the absence of horizontal overlap (and select transistor **123** V.sub.t properties) of one (1) of the tenth V.sub.t-enhanced sections **118**J within the second pillar structure **122**B, due to the impact thereof on the characteristics (e.g., channel types) of the select transistors **123** at the intersections of the second pillar structure **122**B and the first SGP tier **110**K and second SGP tier **110**L. The third pillar structure **122**C may be identified by the absence of horizontal overlap of the ninth V.sub.t-enhanced sections **118**I and the tenth V.sub.t-enhanced sections **118**J with the third pillar structure **122**C, due to the impact thereof on the characteristics (e.g., channel types) of the select transistors **123** at the intersections of the third pillar structure **122**C and the first SGP tier **110**K and second SGP tier **110**L. The fourth pillar structure **122**D may be identified by the horizontal overlap of one (1) of the tenth V.sub.t-enhanced sections **118**J with the second pillar structure **122**B, along with the absence of horizontal overlap of one (1) of the ninth V.sub.t-enhanced sections **118**I with the second pillar structure **122**B, due to the impact thereof on the characteristics (e.g., channel types) of the select transistors **123** at the intersections of the fourth pillar structure **122**D and the first SGP tier **110**K and second SGP tier **110**L.

(93) A non-limiting example of SGPGM operations to set V.sub.t levels for different select transistors **123** within the SGD tier section **111**B and the sense node tier section **111**C of the stack structure **104** is described below by and with reference to Table 1 and Table 2.

(94) TABLE-US-00001 TABLE 1 SB L K J I H G F E D C J C + B L H Pgm H H H H H H H I B + A L H H Pgm H H H H H H H By SB* L/H L/H L/H L/H Pgm H H H H H G By SB* L/H L/H L/H L/H Inh Pgm Pass Pass Pass Pass F By SB* L/H L/H L/H L/H Inh Pass Pgm Pass Pass Pass E By SB* L/H L/H L/H L/H Inh Pass Pass Pgm Pass Pass D By SB* L/H L/H L/H L/H Inh Pass Pass Pass Pgm Pass C By SB* L/H L/H L/H L/H Inh Pass Pass Pass Pass Pgm

(95) TABLE-US-00002 TABLE 2 SB L K J I H, G, F, E, A H H L L D, C B L H H L C L L H H D H L L H

(96) In Table 1, the farthest column to the left represents the tier **110** of the stack structure **104** of the microelectronic device structure **400** in which select transistors **123** are being programmed, wherein J, I, H, G, F, E, D, and C represent programming operations for select transistors **123**

within the second SGP bar tier **110**J, the first SGP bar tier **110**I, the read-amplification SGD tier **110**H, the second SGD tier **110**G, the first SGD tier **110**F, the second SGD bar tier **110**E, the first SGD bar tier **110**D, and the program-inhibit SGD tier **110**C, respectively. In addition, in the uppermost row in Table 1, SB, L, K, J, I, H, G, F, E, D, and C represent the sub-blocks **114** being acted upon, the second SGP tier **110**L, the first SGP tier **110**K, the second SGP bar tier **110**J, the first SGP bar tier **110**I, the read-amplification SGD tier **110**H, the second SGD tier **110**G, the first SGD tier **110**F, the second SGD bar tier **110**E, the first SGD bar tier **110**D, and the program-inhibit SGD tier **110**C, respectively. Furthermore, within the "SB" (sub-block) column, "C+B" represents the combination of a third sub-block **114**C and a second sub-block **114**B of the stack structure **104**; "B+A" represents the combination of a second sub-block **114**B and a first sub-block **114**A of the stack structure **104**; and "By SB*" represents a sub-block selection scheme further described by and with reference to Table 2. In addition, at intersections of the columns and the rows of Table 1, "L" represents a LOW drive state, "H" represents HIGH drive state; "L/H" represents a LOW drive state or a HIGH drive state, according to sub-block address; "Pgm" represents an fulfilled program action for associated select transistors **123**; "Inh" represents a program inhibit action; and "Pass" represents a pass through action.

(97) In Table 2, the farthest column to the left represents the tiers **110** of the stack structure **104** of the microelectronic device structure **400** subject to the "By SB*" sub-block selection scheme identified in Table 1, wherein H, G, F, E, D, and C represent operations for select transistors **123** within the read-amplification SGD tier **110**H, the second SGD tier **110**G, the first SGD tier **110**F, the second SGD bar tier **110**E, the first SGD bar tier **110**D, and the program-inhibit SGD tier **110**C, respectively. In addition, in the uppermost row in Table 2, SB, L, K, J, and I represent the sub-blocks **114** being acted upon, the second SGP tier **110**L, the first SGP tier **110**K, the second SGP bar tier **110**J, and the first SGP bar tier **110**I. Furthermore, A, B, C, and D within the "SB" (sub-block) column represent the first sub-block **114**A, the second sub-block **114**B, the third sub-block **114**C, and the fourth sub-block **114**D, respectively. In addition, at intersections of the columns and the rows of Table 2, "L" represents a LOW drive state, and "H" represents HIGH drive state.

(98) As a non-limiting example, if it is desired to program select transistors **123** at the vertical elevation of the second SGP bar tier **110**J to desired V.sub.t levels, 0 volts (V) may be supplied by a digit line structure **158** to the pillar structures **122** within an individual pillar group **124** by way of a horizontal sense transistor **136** while also biasing the conductive structure **106** of the second SGP bar tier **110**J to a relatively HIGH (H) programming voltage (V.sub.pgm) to facilitate a V.sub.t potential difference. However, if the conductive structure **106** of the second SGP tier **110**L is biased to 0 V, the 0 V supplied by the digit line structure **158** cannot be transferred to the tiers **110** thereunder. Accordingly, the select transistors **123** at intersections of the second SGP tier **110**L and the first pillar structures **122**A within the first sub-block **114**A and the fourth pillar structures **122**D within the fourth sub-block **114**D may be set to an OFF (e.g., inactive) state. Through the combination of 0 V supplied from the digit line structure **158**, the relatively HIGH V.sub.pgm applied to the conductive structure **106** of the second SGP bar tier **110**J, and the OFF state of the select transistors **123** associated with the first pillar structures **122**A and the second pillar structures **122**B at the vertical elevation of the second SGP tier **110**L, select transistors **123** at intersections of the second SGP bar tier **110**J and the second pillar structures **122**B within the second sub-block **114**B and the third pillar structures **122**C within the third sub-block **114**C be programmed to desired V.sub.t levels. In addition, since the select transistors **123** at intersections of the second SGP bar tier **110**J and the first pillar structures **122**A within the first sub-block **114**A and the fourth pillar structures **122**D within the fourth sub-block **114**D have not been programmed through the foregoing process, the second SGP tier **110**L may be utilized in the reverse (e.g., select transistors **123** at intersections of the second SGP tier **110**L and the first pillar structures **122**A within the first sub-block **114**A and the fourth pillar structures **122**D within the fourth sub-block **114**D may be set to an ON state, while select transistors **123** at intersections of the second SGP tier **110**L and the

second pillar structures **122**B within the second sub-block **114**B and the third pillar structures **122**C within the third sub-block **114**C may be set to an OFF state) to program the select transistors **123** at intersections of the second SGP bar tier **110**J and the first pillar structures **122**A within the first sub-block **114**A and the fourth pillar structures **122**D within the fourth sub-block **114**D. The second SGP bar tier **110**J may be employed to program V.sub.t levels of the select transistors **123** at the vertical elevation of the second SGP tier **110**L in a similar manner. In addition, the first SGP bar tier **110**I and the first SGP tier **110**K may be utilized in combination with one another to program V.sub.t levels of the select transistors **123** and the vertical elevations thereof in a similar manner. Once the select transistors **123** within the sense node tier section **111**C of the stack structure **104** are programmed to desired V.sub.t levels, the pillar structures **122** may be individually accessed, and V.sub.t levels of select transistors **123** within tiers **110** of the stack structure **104** underlying the sense node tier section **111**C may be programmed as desired.

(99) During use and operation of a microelectronic device (e.g., a memory device, such as a 3D NAND Flash memory device) including the microelectronic device structure **100**, memory cell programming operations may also be performed. Such operations are referred to herein as "program operations." As a non-limiting example, still referring to FIG. **1**A, if it is desired to program a memory cell **121** operatively associated with the second pillar structure **122**B within the second sub-block **114**B in the sub-section B.sub.1 of the microelectronic device structure **100**, the conductive structures **106** of the second SGD tier **110**G and the first SGD bar tier **110**D may be grounded, and the conductive structures **106** of the GG tier **110**M, the first SGD tier **110**F, the second SGD bar tier **110**E, and the read-amplification SGD tier **110**H may be biased HIGH. As a result, the second transistors **123** at vertical elevations of the second SGD tier **110**G, the first SGD bar tier **110**D, the first SGD tier **110**F, and the second SGD bar tier **110**E may facilitate access to second pillar structure **122**B within the second sub-block **114**B alone. In addition, the conductive structure **106** of the program-inhibit SGD tier **110**C may be biased to a program-inhibit voltage (V.sub.SGD_Inh). Depending on digit line size potential, if the digit line structure **158** supplies 0 V bias to the vertical position of the second SGD bar tier **110**E, back-bias effect may effectuate the discharge of the second pillar structure **122**B. Thus, the second pillar structure **122**B may be biased to 0 V, and if the conductive structure **106** of an active access line tier **110**A intersecting the second pillar structure **122**B is being driven to V.sub.pgm, the memory cell **121** defined at the intersection will be programmed. Conversely, if the digit line structure **158** is biased to supply voltage (V.sub.cc), the select transistor **123** at the intersection of the second pillar structure **122**B and the program-inhibit SGD tier **110**C is not activated due to relatively higher source side potential, and memory cell **121** programming is inhibited.

(100) During use and operation of a microelectronic device (e.g., a memory device, such as a 3D NAND Flash memory device) including the microelectronic device structure **100**, memory cell read operations may also be performed. Such operations are referred to herein as "read operations." As a non-limiting example, during a read operation, write selector transistors **131** at the vertical elevation of the first selector tier **138**A (e.g., employed as a write selector tier) may be set to an OFF state and a flow of current may be directed along a path extending from a source line structure **152**, through each of a first read select pillar structure **146**, a horizontal sense transistor **136**, a second read select pillar structure **148**, a local strap structure **154**, and a digit line structure **158**, and to a digit line structure **158**. During the read operation, the horizontal sense transistor **136** may be used to sense the potential of the SGD plug structure **126**, and the potential of the sense node **112** may change (e.g., from 4 V to 0 V) depending on the status of selected memory cells **121**. The conductive structures **106** of all of the tiers **110** (e.g., the first SGP bar tier **110**I, the second SGP bar tier **110**J, the first SGP tier **110**K, the second SGP tier **110**L, and the GG tier **110**M) within the sense node tier section **111**C may be pre-charged to a desired level, such as 4 V (without limitation), and the select transistors **123** at vertical positions thereof may be provided in an ON state. Accordingly, the sense node **112** may act as a capacitive node having a desired potential, such

as 4 V (without limitation). If, for example, it is desired to read a memory cell **121** operatively associated with a second pillar structure **122**B within the second sub-block **114**B in the sub-section B.sub.1 of the microelectronic device structure **100**, the conductive structures **106** of the second SGD tier **110**G and the first SGD bar tier **110**D may be grounded and conductive structures **106** of the first SGD tier **110**F and the second SGD bar tier **110**E may be biased HIGH. By doing so, the select transistors **123** at vertical elevations of the second SGD tier **110**G, the first SGD bar tier **110**D, the first SGD tier **110**F, and the second SGD bar tier **110**E may facilitate access to the second pillar structure **122**B within the second sub-block **114**B alone. In addition, the conductive structure **106** of the read-amplification SGD tier **110**H may be utilized to provide a bias, and the select transistor **123** defined at the intersection of the second pillar structure **122**B and the read-amplification SGD tier **110**H may be employed as a clamp transistor and may facilitate a back bias effect. A lower portion of the second pillar structure **122**B may be pre-charged to a desired level, such as 1 V (without limitation). If 1 V reaches the vertical elevation of the second SGD tier **110**G, the relatively higher potential below the read-amplification SGD tier **110**H results in an OFF state of the select transistor **123** at the intersection of the second pillar structure **122**B and the read-amplification SGD tier **110**H and the 4 V potential of the sense node **112** is maintained. However, once the memory cell **121** associated with the second pillar structure **122**B is ramped down to 0 V, depending on the status of the associated active access line tier **110**A, the second pillar structure **122**B will discharge. If the access transistor of the read-amplification SGD tier **110**H is in an OFF state, then 1 V is maintained at the top portion of the second pillar structure **122**B, and the select transistor **123** at the intersection of the second pillar structure **122**B and the read-amplification SGD tier **110**H is in an OFF state such that the 4 V potential of the sense node **112** is maintained. However, if the access transistor of the read-amplification SGD tier **110**H is in an ON state, then after ramping the source to 0 V, a portion of the second pillar structure **122**B below the read-amplification SGD tier **110**H will be discharged to close to 0 V, resulting in an ON state of the select transistor **123** at the intersection of the second pillar structure **122**B and the read-amplification SGD tier **110**H to leak charge from the sense node **112** and bring the potential thereof to close to 0 V. By sensing the potential difference between 0 V and 4 V, horizontal sense transistor **136** facilitates read out the data stored in the memory cell **121**.

(101) Having pillar structures **122** within multiple different sub-blocks **114** share an individual sense node **112** with one another provides efficiency and performance benefits relative to conventional configurations wherein sensing circuitry is simply shared by pillar structures within the same sub-block as one another as it alleviates a need for sequential reading of all sub-blocks **114** to access specific sub-block data. In addition, the continuity of the conductive structures **106** of the tiers **110** within the sense node tier section **111**C and the stacked SGD tier section **111**B across and between different sub-blocks **114** of an individual block **164** alleviates the need for the complex and costly patterning conventionally required to form segmented SGD structures (e.g., discrete SGD structures, each discrete SGD structure within a different sub-block than each other discrete SGD structure) within an individual SGD tier, as well as undesirable resistivity characteristics (e.g., relatively high resistivity) that may be associated with segmented SGD structures within an individual block. Furthermore, as sense node capacitance is orders of magnitude lower than digit line capacitance, discharge can be achieved using a relatively small amount of current, promoting relatively faster sense and read operations. Moreover, as the current path through a horizontal sense transistor **136** may be shorter than a current path through relatively long pillar structure **122**, current drive rating may be relatively higher and may facilitate faster read operation and simpler page buffer design as compared to conventional configurations.

(102) Thus, a microelectronic device according to embodiments of the disclosure comprises a microelectronic device comprises a stack structure, pillar structures, a conductive plug structure, a sense transistor, and selector transistors. The stack structure comprises a vertically alternating sequence of conductive material and insulative material. The stack structure is divided into blocks

separated by dielectric slot structures. The blocks individually include sub-blocks horizontally extending in parallel with one another. The pillar structures vertically extend through one of the blocks of the stack structure. Each pillar structure of a group of the pillar structures is horizontally positioned within a different one of the sub-blocks of the one of the blocks than each other pillar structure of the group of the pillar structures. The conductive plug structure is coupled to and horizontally extends across and between multiple of the pillar structures of the group of the pillar structures. The sense transistor is gated by the conductive plug structure. The selector transistors couple the sense transistor to a read source line structure and a digit line structure.

(103) Furthermore, a memory device according to embodiments of the disclosure comprises a stack structure divided into blocks separated by dielectric slot structures, pillar groups within horizontal areas of the blocks, horizontal sense transistors vertically overlying and coupled to the pillar groups, vertical read selector transistors vertically overlying and coupled to the horizontal sense transistors, vertical write selector transistors vertically overlying and coupled to the horizontal sense transistors, and vertical write selector transistors vertically overlying and coupled to the horizontal sense transistors. Each of the each of the block of the stack structure comprises an access line section comprising tiers including access line structures, a select gate section overlying the access line section and comprising additional tiers including drain side select gate (SGD) structures, and a sense node section overlying the select gate section and comprising further tiers including select gate programming (SGP) structures. The pillar groups individually comprise multiple pillar structures vertically extending completely through one of the blocks, each pillar structure of the multiple pillar structures horizontally positioned within a different sub-block of the one of the blocks than each other pillar structure of the multiple pillar structures. The vertical write selector transistors are horizontally offset from the vertical read selector transistors.

(104) In additional embodiments, the microelectronic device structure **100** may be formed to have a different configuration than that previously described with reference to FIGS. **1**A and **1**B. The microelectronic device structure **100** may, for example, be formed to exhibit a configuration such as one of the configurations depicted in FIGS. **2** and **3** and described in further detail below. With the description provided below, it will be readily apparent to one of ordinary skill in the art that the structures and devices described herein may be included in relatively larger structures, devices, and systems.

(105) Before referring to FIG. **2**, it will be understood that throughout FIGS. **2**, **3**, and **4**A through **4**M and the associated description, features (e.g., regions, materials, structures, devices) functionally similar previously described features (e.g., previously described materials, structures, devices) are referred to with similar reference numerals incremented by 100. To avoid repetition, not all features shown in FIGS. **2**, **3**, and **4**A through **4**M are described in detail herein. Rather, unless described otherwise below, a feature in one or more of FIGS. **2**, **3**, and **4**A through **4**M designated by a reference numeral that is a 100 increment of the reference numeral of a feature previously described with reference to one or more of FIGS. **1**A and **1**B will be understood to be substantially similar to the previously described feature. As a non-limiting example, unless described otherwise below, features designated by the reference numerals **236**, **336**, and **436** in FIGS. **2**, **3**, and **4**F (and subsequent FIGS. **4**G through **4**M), respectively, will be understood to respectively be substantially similar to the horizontal sense transistors **136** previously described herein with reference to FIGS. **1**A and **1**B.

(106) FIG. **2** is simplified, partial top-down view of a microelectronic device structure **200** for a microelectronic device (e.g., a memory device, such as a NAND Flash memory device), in accordance with additional embodiments of the disclosure. The microelectronic device structure **200** may be similar to the microelectronic device structure **100** previously described with reference to FIGS. **1**A and **1**B, except that, for example, relative horizontal orientations and/or relative horizontal positions of some features of the microelectronic device structure **200** within horizontal areas of individual blocks **264** of a stack structure **204** are different than relative horizontal

orientations and/or relative horizontal positions of corresponding features of the microelectronic device structure **100**. For instance, as shown in FIG. **2**, horizontal orientations of SGD plug structures **226** within a horizontal area of an individual block **264** of the stack structure **204** may be different than horizontal orientations of the SGD plug structures **126** of the microelectronic device structure **100** previously described with reference to FIG. **1**B. As a result, horizontal orientations and/or horizontal positions of additional features of the microelectronic device structure **200** operatively associated with the SGD plug structures **226** may be different than horizontal orientations and/or horizontal positions of additional features of the microelectronic device structure **100** operatively associated with the SGD plug structures **126**, as described in further detail below. It will be understood that a simplified, partial longitudinal cross-sectional view of the microelectronic device structure **200** about dashed line A.sub.2-A.sub.2 depicted in FIG. **2** is substantially similar to the simplified, partial longitudinal cross-sectional view of the microelectronic device structure **100** shown in FIG. **1**A.
(107) As shown in FIG. **2**, the SGD plug structures **226** of the microelectronic device structure **200** may individually horizontally extend in the X-direction. Within a horizontal area of an individual block **264** of the stack structure **204**, the SGD plug structures **226** may horizontally extend in parallel in the X-direction with the slot structures **266** horizontally neighboring the block **264** in the Y-direction; and may horizontally extend perpendicular to the digit line structures **258** horizontally extending in the Y-direction. Some of the SGD plug structures **226** may horizontally extend in parallel with one another in the X-direction; and other of the SGD plug structures **226** may horizontally extend in series with one another in the X-direction.
(108) Similar to the SGD plug structures **126** (FIGS. **1**A and **1**B), each of the SGD plug structures **226** may individually be operatively associated with a pillar group including four (4) pillar structures **222** vertically extending through the stack structure **204**. An individual SGD plug structure **226** may be coupled to and may horizontally extend in the X-direction across and between the four (4) of the pillar structures **222** of the pillar group associated therewith. The four (4) pillar structures **222** of an individual pillar group operatively associated with an individual SGD plug structure **226** may substantially horizontally aligned with one another in the Y-direction.
(109) Still referring to FIG. **2**, each block **264** of the stack structure **204** may be sub-divided into multiple (e.g., a plurality of, more than one) sub-blocks **214**. The sub-blocks **214** may horizontally extend parallel with one another in a direction acutely angled relative to each of the X-direction and the Y-direction shown in FIG. **2**. Each of the sub-blocks **214** may individually be operatively associated with at least one set of the pillar structures **222** horizontally extending in the direction acutely angled relative to each of the X-direction and the Y-direction. In addition, for an individual block **264**, multiple (e.g., a plurality of, more than one) sub-blocks **214** thereof may be grouped together with one another within sub-block groups **216**. In FIG. **2**, two (2) sub-block groups **216** are depicted, a second sub-block group **216**B and a third sub-block group **216**C. Each of the sub-block groups **216** may individually include a first sub-block **214**A, a second sub-block **214**B, a third sub-block **214**C, and a fourth sub-block **214**D. Sub-block groups **216** horizontally neighboring one another within an individual block **264** may exhibit an inverse horizontal arrangement of the different sub-blocks **214** thereof relative to one another. For example, the first sub-block **214**A of the second sub-block group **216**B may be most horizontally proximate the first sub-block **214**A of the third sub-block group **216**C horizontally neighboring the second sub-block group **216**B.
(110) Similar to the configuration of the microelectronic device structure **100** previously described with reference to FIGS. **1**A and **1**B, pillar structures **222** of the microelectronic device structure **200** operatively associated with an individual SGD plug structure **226** may be located within different sub-blocks **214** of an individual block **264** than one another. For example, within the horizontal area of a sub-section B.sub.2 of the microelectronic device structure **200**, one (1) of the SGD plug structures **226** may vertically overlie and be coupled to each of four (4) pillar structures **222** located

in different sub-blocks **214** within the third sub-block group **216**C than one another. A first of the pillar structures **222** may be located within the first sub-block **214**A of the third sub-block group **216**C; a second of the pillar structure **222** may be located within the second sub-block **214**B of the third sub-block group **216**C; a third of the pillar structures **222**C may be located within the third sub-block **214**C of the third sub-block group **216**C; and a fourth of the pillar structure **222** may be located within the fourth sub-block **214**D of the third sub-block group **216**C.

(111) With continued reference to FIG. **2**, channel structures **234** of the microelectronic device structure **200** may be positioned within horizontal areas of the SGD plug structures **226**, and may individually horizontally extend in the X-direction. The SGD plug structures **226**, the channel structures **234**, and gate dielectric structures (e.g., corresponding to the gate dielectric structures **130** previously described with reference to FIG. **1**A) may together form horizontal sense transistors **236** substantially similar to the horizontal sense transistors **136** previously described with reference to FIGS. **1**A and **1**B, except for the horizontal orientations thereof. In addition, the microelectronic device structure **200** may include tiers, memory cells, select transistors, V.sub.t-enhanced sections, sense nodes, write pillar structures **244**, first read pillar structures **246**, second read pillar structures **248**, selector tiers, doped sections, write selector transistors, read selector transistors, read source line structures, local strap structures **254**, digit line contact structures **260**, digit line structures **258**, and isolation material respectively substantially similar to and utilized in substantially the same manner as the tiers **110**, the memory cells **121**, the select transistors **123**, the V.sub.t-enhanced sections **118**, the sense nodes **112**, the write pillar structures **144**, the first read pillar structures **146**, the second read select pillar structures **148**, the selector tiers **138**, the doped sections **142**, the write selector transistors **131**, the read selector transistors **135**, the read source line structures **152**, the local strap structures **154**, the digit line contact structures **160**, the digit line structures **158**, and the isolation material **162** previously described with reference to FIGS. **1**A and **1**B.

(112) FIG. **3** is simplified, partial longitudinal cross-sectional view of a microelectronic device structure **300** for a microelectronic device (e.g., a memory device, such as a NAND Flash memory device), in accordance with further embodiments of the disclosure. The microelectronic device structure **300** may be similar to the microelectronic device structure **100** previously described with reference to FIGS. **1**A and **1**B, except that, for example, the microelectronic device structure **300** may have a reduced quantity of tiers **310** and/or some of the tiers **310** may be utilized differently for various operations (e.g., SGPGM operations, program operations, read operations, erase operations) of a microelectronic device (e.g., a memory device, such as a 3D NAND Flash memory device) including the microelectronic device structure **300**. For instance, as shown in FIG. **3**, the microelectronic device structure **300** may be free of (e.g., may not include) tiers **310** corresponding to the read-amplification SGD tier **110**H, the first SGP bar tier **110**I, the second SGP bar tier **110**J, the first SGP tier **110**K, and the second SGP tier **110**L of the microelectronic device structure **100** previously described with reference to FIG. **1**A. Accordingly, a sense node tier section **311**C of a stack structure **304** of the microelectronic device structure **300** may modified relative to the sense node tier section **111**C (FIG. **1**A) of the microelectronic device structure **100** (FIG. **1**A), and may include one or more (e.g., two or more, three or more) GG tiers **310**M. Sense nodes **312** of the microelectronic device structure **300** may be defined by and include SGD plug structures **326** and portions of the GG tiers **310**M. The sense nodes **312** may not include portions of SGP bar tiers and SGP tiers corresponding to the first SGP bar tier **110**I, the second SGP bar tier **110**J, the first SGP tier **110**K, and the second SGP tier **110**L of the microelectronic device structure **100** previously described with reference to FIG. **1**A.

(113) In some embodiments, the microelectronic device structure **300** includes multiple (e.g., two or more, three or more) of the GG tiers **310**M. For example, as shown in FIG. **3**, the stack structure **304** of the microelectronic device structure **300** may include three (3) of the GG tiers **310**M in a vertically stacked arrangement with one another. The GG tiers **310**M may be vertically interposed between a second SGD tier **310**G of the stack structure **304** and a SGD plug tier **327** of the

microelectronic device structure **300**.

(114) Since the microelectronic device structure **300** does not include tiers **310** corresponding to the read-amplification SGD tier **110**H, the first SGP bar tier **110**I, the second SGP bar tier **110**J, the first SGP tier **110**K, and the second SGP tier **110**L of the microelectronic device structure **100** previously described with reference to FIG. **1**A, operations (e.g., SGPGM operations, program operations, read operations, erase operations) that may otherwise utilize such omitted (e.g., absent) tiers **310** (e.g., in the manners previously described in relation to use and operation of a microelectronic device including the microelectronic device structure **100**) may be effectuated using different tiers **310** of the microelectronic device structure **300**. For example, for program operations, the first SGD bar tier **110**D and the second SGD bar tier **110**E may be employed to program V.sub.t levels of the first SGD tier **110**F and the second SGD tier **110**G, respectively; and the first SGD bar tier **110**D, the second SGD bar tier **110**E, the first SGD tier **110**F, and the second SGD tier **110**G may be employed to program V.sub.t levels of the program-inhibit SGD tier **310**C. In addition, for read operations, the program-inhibit SGD tier **310**C in a manner similar to that of the read-amplification SGD tier **110**H previously described with reference to FIGS. **1**A and **1**B; and the GG tiers **310**M may all be provided in an ON state, to effectively serve as a coupling capacitor for the sense nodes **312**.

(115) FIGS. **4**A through **4**M are simplified, partial longitudinal cross-sectional views illustrating a microelectronic device structure at different processing stages of a method of forming a microelectronic device (e.g., a memory device, such as a 3D NAND Flash memory device), in accordance with embodiments of the disclosure. The methods described herein with reference to FIGS. **4**A through **4**M may be used to form one or more of the microelectronic device structures (e.g., the microelectronic device structure **100**, the microelectronic device structure **200**, the microelectronic device structure **300**) of the disclosure previously described herein with reference to FIGS. **1**A and **1**B, **2**, and **3**. With the description provided below, it will be readily apparent to one of ordinary skill in the art that the methods described herein may be used for forming various devices. In other words, the methods of the disclosure may be used whenever it is desired to form a microelectronic device.

(116) FIG. **4**A is a simplified, longitudinal cross-sectional view of a portion of a microelectronic device structure **400** at a processing stage of a method of forming a microelectronic device, in accordance with embodiments of the disclosure. A horizontal position of the portion of the microelectronic device structure **400** shown in FIG. **4**A may correspond to the horizontal position of the sub-section B.sub.1 of the microelectronic device structure **100** previously described with reference to FIGS. **1**A and **1**B; and a vertical position of the portion of the microelectronic device structure **400** shown in FIG. **4**A may correspond to that of a portion of the microelectronic device structure **100** vertically extending between the first SGP tier **110**K (FIG. **1**A) of the stack structure **104** (FIG. **1**A) and an upper vertical boundary of the stack structure **104** (FIG. **1**A).

(117) As shown in FIG. **4**A, the microelectronic device structure **400** may be formed to include a preliminary stack structure including a vertically alternating sequence of insulative material **405** and sacrificial material **407** arranged in preliminary tiers **409**. The sacrificial material **407** may be vertically interleaved with the insulative material **405**, and each of the preliminary tiers **409** may include the sacrificial material **407** vertically neighboring the insulative material **405**. In addition, the microelectronic device structure **400** may be formed to include pillar structures **422** vertically extending through the preliminary tiers **409** of the preliminary stack structure. The pillar structures **422** may correspond to the pillar structures **122** previously described herein with reference to FIGS. **1**A and **1**B.

(118) The sacrificial material **407** of each of the preliminary tiers **409** of the preliminary stack structure may be formed of and include at least one material (e.g., at least one insulative material) that may be selectively removed relative to the insulative material **405** during subsequent processing (e.g., subsequent “replacement gate” or “gate last” processing) of the microelectronic

device structure **400**, as described in further detail below. A material composition of the sacrificial material **407** is different than a material composition of the insulative material **405**. The sacrificial material **407** may be selectively etchable relative to the insulative material **405** during common (e.g., collective, mutual) exposure to a first etchant, and the insulative material **405** may be selectively etchable to the sacrificial material **407** during common exposure to a second, different etchant. As a non-limiting example, the sacrificial material **407** may be formed of and include additional insulative material, such as one or more of at least one dielectric oxide material (e.g., one or more of SiO.sub.x, phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass, AlO.sub.x, HfO.sub.x, NbO.sub.x, TiO.sub.x, ZrO.sub.x, TaO.sub.x, and MgO.sub.x), at least one dielectric nitride material (e.g., SiN.sub.y), at least one dielectric oxynitride material (e.g., SiO.sub.xN.sub.y), and at least one dielectric carboxynitride material (e.g., SiO.sub.xC.sub.zN.sub.y). In some embodiments, the sacrificial material **407** is formed of and includes dielectric nitride material, such as SiN.sub.y (e.g., Si.sub.3N.sub.4). The sacrificial material **407** of each of the preliminary tiers **409** may be substantially homogeneous, or the sacrificial material **407** of one or more (e.g., each) of the preliminary tiers **409** may be heterogeneous.

(119) The insulative material **405** of each of the preliminary tiers **409** of the preliminary stack structure may be formed of and include at least one dielectric material, such one or more of at least one dielectric oxide material (e.g., one or more of SiO.sub.x, phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass, AlO.sub.x, HfO.sub.x, NbO.sub.x, TiO.sub.x, ZrO.sub.x, TaO.sub.x, and MgO.sub.x), at least one dielectric nitride material (e.g., SiN.sub.y), at least one dielectric oxynitride material (e.g., SiO.sub.xN.sub.y), and at least one dielectric carboxynitride material (e.g., SiO.sub.xC.sub.zN.sub.y). In some embodiments, the insulative material **405** of each of the preliminary tiers **409** of the preliminary stack structure is formed of and includes dielectric oxide material, such as SiO.sub.x (e.g., SiO.sub.2). The insulative material **405** of each of the preliminary tiers **409** may be substantially homogeneous, or the insulative material **405** of one or more (e.g., each) of the preliminary tiers **409** may be heterogeneous.

(120) The pillar structures **422** may each individually be formed of and include a stack of materials facilitating the use of the pillar structures **422** to form vertically extending strings of memory cells following subsequent processing acts, as described in further detail below. By way of non-limiting example, each of the pillar structures **422** may be formed to include cell film material **419**A, channel material **419**B, and dielectric fill material **419**C. The cell film material **419**A may be formed on or over surfaces of the sacrificial material **407** and the insulative material **405** of the preliminary tiers **409** of the preliminary stack structure and may be formed of and include a stack of materials, such as a charge-blocking material (e.g., a first dielectric oxide material, such as one or more of SiO.sub.x, and AlO.sub.x), a charge-trapping material (e.g., dielectric nitride material, such as SiN.sub.y) on the charge-blocking material, and a gate dielectric material (e.g., a second dielectric oxide material, such as SiO.sub.x) on the charge-trapping material. The channel material **419**B may located on the gate dielectric material of the cell film material **419**A, and may be formed of and include semiconductive material (e.g., silicon, such as polycrystalline silicon). The dielectric fill material **419**C may located on the channel material **419**B, and may be formed of and include dielectric material (e.g., one or more of dielectric oxide material, dielectric nitride material, and an air gap).

(121) As shown in FIG. **4**A, in some embodiments, the cell film material **419**A and the channel material **419**B are formed to substantially continuously extend (e.g., horizontally extend, vertically extend) from and between horizontally neighboring pillar structures **422** within horizontal areas of different sub-blocks **414** (e.g., a first sub-block **414**A, a second sub-block **414**B, a third sub-block **414**C, a fourth sub-block **414**D) within the horizontal area of an individual sub-block group **416** (e.g., a third sub-block group **416**C). For example, portions of the cell film material **419**A and the channel material **419**B may substantially continuously horizontally extend on or over an uppermost

surface of the preliminary stack structure (e.g., an uppermost surface of an uppermost preliminary tier **409**). In additional embodiments, the cell film material **419**A and the channel material **419**B do not horizontally extend from and between horizontally neighboring pillar structures **422** within the horizontal areas of horizontally neighboring sub-blocks **414**. For example, the cell film material **419**A and the channel material **419**B forming a portion of individual pillar structure **422** may be substantially confined within a horizontal area of the sub-block **414** in which the pillar structure **422** is located.

(122) Still referring to FIG. **4**A, for each pillar structure **422**, an upper vertical boundary (e.g., an upper surface) of the dielectric fill material **419**C thereof may be formed to be vertically recessed (e.g., to vertically underlie) relative to an upper vertical boundary of the channel material **419**B thereof. Accordingly, inner side surfaces (e.g., inner sidewalls) of the channel material **419**B forming portions of the pillar structures **422** may be exposed by openings (e.g., plug recesses) vertically extending into upper regions of the pillar structures **422**.

(123) Referring next to FIG. **4**B, which is a simplified, longitudinal cross-sectional view of the portion of the microelectronic device structure **400** shown in FIG. **4**A following the processing stage previously described with reference to FIG. **4**A, at least one first masking structure **466** including openings **468** vertically extending therethrough may be formed on or over an uppermost one of the preliminary tiers **409** of the preliminary stack structure, and then one or more conductivity-enhancing species may be provided (e.g., implanted) into one or more of the preliminary tiers **409** at horizontal positions of the openings **468** in the first masking structure **466** to form one or more V.sub.t-enhanced sections **418** of the microelectronic device structure **400**. As a non-limiting example, as shown in FIG. **4**B, ninth V.sub.t-enhanced sections **4181** corresponding to the ninth V.sub.t-enhanced sections **118**I previously described with reference to FIG. **1**A may be formed at horizontal positions and vertical locations corresponding to those previously described with reference to FIG. **1**A.

(124) In some embodiments, the first masking structure **466** is a photoresist mask formed of and including photoresist material, such as positive tone photoresist material or negative tone photoresist material. Suitable photoresist materials (e.g., positive tone photoresist materials, negative tone photoresist materials) are known in the art, and are, therefore, not described in detail herein. The first masking structure **466** may, for example, be compatible with 13.7 nm, 157 nm, 193 nm, 248 nm, or 365 nm wavelength systems; with 193 nm wavelength immersion systems; and/or with electron beam lithographic systems. In addition, the openings **468** in the first masking structure **466** may be formed using conventional processes (e.g., conventional photolithographic patterning and development processes) and conventional equipment, which are not described in detail herein.

(125) In some embodiments, at least one P-type dopant (e.g., one or more of B, Al, and Ga) is implanted at one or more vertical positions (e.g., at a vertical position corresponding to that of first SGP tier **110**K previously described with reference to FIG. **1**A) within the microelectronic device structure **400** at the processing stage of FIG. **4**B to form some of the V.sub.t-enhanced sections **418** of the microelectronic device structure **400**. The conductivity-enhancing species (e.g., P-type dopant) may be implanted into the microelectronic device structure **400** using conventional implantation processes (e.g., a plasma doping (PLAD) implantation process, a beam-line (BL) implantation process) and equipment, which are not described in detail herein.

(126) Referring next to FIG. **4**C, which is a simplified, longitudinal cross-sectional view of the portion of the microelectronic device structure **400** shown in FIG. **4**B following the processing stage previously described with reference to FIG. **4**B, the first masking structure **466** may be removed and an second masking structure **470** include additional openings **472** vertically extending therethrough may be formed on or over the uppermost one of the preliminary tiers **409** of the preliminary stack structure. Horizontal positions of the additional openings **472** in the second masking structure **470** may be at least partially horizontally offset from the horizontal positions of

the openings **468** (FIG. **4**B) in the first masking structure **466** (FIG. **4**B). Following the formation of the second masking structure **470**, one or more conductivity-enhancing species may be provided (e.g., implanted) into one or more other of the preliminary tiers **409** at the horizontal positions of the additional openings **472** in the second masking structure **470** to form more of the V.sub.t-enhanced sections **418** of the microelectronic device structure **400**. As a non-limiting example, as shown in FIG. **4**C, tenth V.sub.t-enhanced sections **418**J corresponding to the tenth V.sub.t-enhanced sections **118**J previously described with reference to FIG. **1**A may be formed at horizontal positions and vertical locations corresponding to those previously described with reference to FIG. **1**A.

(127) In some embodiments, the second masking structure **470** is an additional photoresist mask formed of and including additional photoresist material, such as additional positive tone photoresist material or additional negative tone photoresist material. The second masking structure **470** may, for example, be compatible with 13.7 nm, 157 nm, 193 nm, 248 nm, or 365 nm wavelength systems; with 193 nm wavelength immersion systems; and/or with electron beam lithographic systems. Furthermore, the additional openings **472** in the second masking structure **470** may be formed using conventional processes (e.g., conventional photolithographic patterning and development processes) and conventional equipment, which are not described in detail herein.

(128) In some embodiments, at least one P-type dopant (e.g., one or more of B, Al, and Ga) is implanted at one or more vertical positions (e.g., at a vertical position corresponding to that of second SGP tier **110**L previously described with reference to FIG. **1**A) within the microelectronic device structure **400** at the processing stage of FIG. **4**C to form some more of the V.sub.t-enhanced sections **418** of the microelectronic device structure **400**. The conductivity-enhancing species (e.g., P-type dopant) may be implanted into the microelectronic device structure **400** using conventional implantation processes (e.g., PLAD implantation process, a BL implantation process) and equipment, which are not described in detail herein.

(129) Referring next to FIG. **4**D, which is a simplified, longitudinal cross-sectional view of the portion of the microelectronic device structure **400** shown in FIG. **4**C following the processing stage previously described with reference to FIG. **4**C, the second masking structure **470** (FIG. **4**C) may be removed, and then SGD plug material **425**, gate dielectric material **429**, channel material **433**, backside dielectric material **473**, and dielectric cap material **475** may be formed in sequence over the pillar structures **422** and the uppermost one of the preliminary tiers **409** of the preliminary stack structure. The SGD plug material **425** may be formed on or over the channel material **419**B and the dielectric fill material **419**C of the pillar structures **422**, and may substantially continuously extend from and between the pillar structures **422**. The gate dielectric material **429** may be formed on or over the SGD plug material **425**. The channel material **433** may be formed on or over the gate dielectric material **429**. The backside dielectric material **473** may be formed on or over the channel material **433**. The dielectric cap material **475** may be formed on or over the backside dielectric material **473**.

(130) As shown in FIG. **4**D, the SGD plug material **425** may be formed to substantially fill the openings (e.g., plug recesses) in the upper regions of the pillar structures **422** partially defined by the upper vertical boundaries of the dielectric fill material **419**C of the pillar structures **422**. Accordingly, portions of the SGD plug material **425** may contact (e.g., physically contact, electrically contact) inner side surfaces (e.g., inner sidewalls) and upper surfaces of the channel material **419**B forming portions of the pillar structures **422**, and may also physically contact upper surfaces of the dielectric fill material **419**C forming portions of the pillar structures **422**. The SGD plug material **425** may vertically overlie the pillar structures **422**, and may horizontally extend across and between the pillar structures **422**. An upper vertical boundary (e.g., an upper surface) of the SGD plug material **425** may be substantially planar, and a lower vertical boundary (e.g., a lower surface) of the SGD plug material **425** may be non-planar. Within a horizontal area of an individual sub-block group **416** (e.g., the third sub-block group **416**C), some of the pillar structures **422**

covered by the SGD plug material **425** may be located in the same sub-block **414** as one another, and other of the pillar structures **422** covered by the SGD plug material **425** may be located in different sub-blocks **414** than one another. A material composition of the SGD plug material **425** may correspond to (e.g., be the same as) the material composition of the SGD plug structure **126** previously described with reference to FIGS. **1**A and **1**B. In some embodiments, the SGD plug material **425** is formed of and includes N-type polycrystalline silicon (e.g., polycrystalline silicon doped with at least one N-type dopant, such as polycrystalline silicon doped with phosphorous (P)).
(131) The gate dielectric material **429** may be formed to vertically overlie and substantially continuously horizontally extend across and cover an upper surface of the SGD plug material **425**. A material composition of the gate dielectric material **429** may correspond to (e.g., be the same as) the material composition of the gate dielectric structures **130** previously described with reference to FIG. **1**A. In some embodiments, the gate dielectric material **429** is formed of and includes SiO.sub.x (e.g., SiO.sub.2).
(132) The channel material **433** may be formed to vertically overlie and substantially continuously horizontally extend across and cover an upper surface of the gate dielectric material **429**. A material composition of the channel material **433** may correspond to (e.g., be the same as) the material composition of the channel structures **134** previously described with reference to FIGS. **1**A and **1**B. In some embodiments, the channel material **433** is formed of and includes lightly P-type doped polycrystalline silicon (e.g., polycrystalline silicon doped with at least one P-type dopant, such as polycrystalline silicon doped with B).
(133) The backside dielectric material **473** may be formed to vertically overlie and substantially continuously horizontally extend across and cover an upper surface of the channel material **433**. The backside dielectric material **473** may be formed of and include dielectric material, such as one or more of at least one dielectric oxide material (e.g., one or more of SiO.sub.x, AlO.sub.x, phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass), at least one dielectric nitride material (e.g., SiN.sub.y), and at least one low-K dielectric material (e.g., one or more of SiO.sub.xC.sub.y, SiO.sub.xN.sub.y, SiC.sub.xO.sub.yH.sub.z, and SiO.sub.xC.sub.zN.sub.y). The backside dielectric material **473** may be substantially homogeneous, or the backside dielectric material **473** may be heterogeneous. In some embodiments, the backside dielectric material **473** is formed of and includes SiO.sub.x (e.g., SiO.sub.2).
(134) The dielectric cap material **475** may be formed to vertically overlie and substantially continuously horizontally extend across and cover an upper surface of the backside dielectric material **473**. The dielectric cap material **475** may be formed of and include dielectric material, such as one or more of at least one dielectric oxide material (e.g., one or more of SiO.sub.x, AlO.sub.x, phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass), at least one dielectric nitride material (e.g., SiN.sub.y), and at least one low-K dielectric material (e.g., one or more of SiO.sub.xC.sub.y, SiO.sub.xN.sub.y, SiC.sub.xO.sub.yH.sub.z, and SiO.sub.xC.sub.zN.sub.y). A material composition of the dielectric cap material **475** may be different than a material composition of the backside dielectric material **473**. The dielectric cap material **475** may be substantially homogeneous, or the dielectric cap material **475** may be heterogeneous. In some embodiments, the dielectric cap material **475** is formed of and includes SiN.sub.y (e.g., Si.sub.3N.sub.4).
(135) Referring next to FIG. **4**E, which is a simplified, longitudinal cross-sectional view of the portion of the microelectronic device structure **400** shown in FIG. **4**D following the processing stage previously described with reference to FIG. **4**D, a third masking structure **478** may be formed on or over the dielectric cap material **475**, and then portions of the dielectric cap material **475**, the backside dielectric material **473**, the channel material **433**, the gate dielectric material **429** (FIG. **4**D), the SGD plug material **425** (FIG. **4**D), and the channel material **419**B. The material removal process may form first openings **480** vertically extending from an upper boundary (e.g., an upper

surface) of a remaining portion of the third masking structure **478** to or beyond an upper boundary of the cell film material **419**A.

(136) The material removal process of FIG. **4**E may form SGD plug structures **426** from (e.g., from remaining portions of) the SGD plug material **425** (FIG. **4**D), and may form gate dielectric structures **430** from (e.g., from remaining portions of) the gate dielectric material **429** (FIG. **4**D). The SGD plug structures **426** may be formed to have configurations corresponding to (e.g., substantially the same as) the configurations of the SGD plug structures **126** previously described with reference to FIGS. **1**A and **1**B. In addition, the gate dielectric structures **430** may be formed to have configurations corresponding to (e.g., substantially the same as) the configurations of the gate dielectric structures **130** previously described with reference to FIG. **1**A. The first openings **480** horizontally intervene between and separate horizontally neighboring SGD plug structures **126** and horizontally neighboring gate dielectric structures **130**.

(137) In some embodiments, the third masking structure **478** is a photoresist mask formed of and including photoresist material, such as positive tone photoresist material or negative tone photoresist material. The third masking structure **478** may, for example, be compatible with 13.7 nm, 157 nm, 193 nm, 248 nm, or 365 nm wavelength systems; with 193 nm wavelength immersion systems; and/or with electron beam lithographic systems. Furthermore, the first openings **480** may be formed using conventional processes (e.g., conventional photolithographic patterning and development processes, conventional etching processes) and conventional equipment, which are not described in detail herein.

(138) Referring next to FIG. **4**F, which is a simplified, longitudinal cross-sectional view of the portion of the microelectronic device structure **400** shown in FIG. **4**E following the processing stage previously described with reference to FIG. **4**E, portions of the third masking structure **478** (FIG. **4**E) may be removed; additional portions of the dielectric cap material **475** (FIG. **4**E), the backside dielectric material **473 475** (FIG. **4**E), the channel material **433 475** (FIG. **4**E) not covered by a remaining portion of the third masking structure **478** (FIG. **4**E) may be removed; and the remaining portion of the third masking structure **478** (FIG. **4**E) may be removed. The material removal process may form second openings **482** from some of the first openings **480** (FIG. **4**E), wherein horizontal dimensions of the second openings **482** at vertical positions above of the gate dielectric structures **430** are relatively greater than horizontal dimensions of the first openings **480** (FIG. **4**E) at the vertical positions above of the gate dielectric structures **430**. As shown in FIG. **4**E, portions of upper surfaces of the gate dielectric structures **430** may be exposed by the second openings **482**.

(139) The material removal process of FIG. **4**F may form channel structures **434** from (e.g., from remaining portions of) the channel material **433** (FIG. **4**E), backside dielectric structures **474** from (e.g., from remaining portions of) the backside dielectric material **473** (FIG. **4**E), and dielectric cap structures **476** from (e.g., from remaining portions of) dielectric cap material **475** (FIG. **4**E). The channel structures **434** may be formed to have configurations corresponding to (e.g., substantially the same as) the configurations of the channel structures **134** previously described with reference to FIGS. **1**A and **1**B. In addition, the backside dielectric structures **474** and the dielectric cap structures **476** may be formed to have horizontal areas, horizontal shape, and horizontal positions substantially the same as the horizontal areas, the horizontal shapes, and the horizontal positions of the channel structures **434**, respectively.

(140) The material removal process of FIG. **4**F may form horizontal sense transistors **436** of the microelectronic device structure **400**. Each of the horizontal sense transistors **436** may individually including one (1) of the SGD plug structures **426**, one (1) of the channel structures **434**, and one (1) of the gate dielectric structures **430** vertically interposed between the one (1) of the SGD plug structures **426** and the one (1) of the channel structures **434**. The horizontal sense transistors **436** may be formed to have configurations corresponding to (e.g., substantially the same as) the configurations of the horizontal sense transistors **136** previously described with reference to FIGS.

1A and **1**B. In addition, each of the horizontal sense transistors **436** may be operatively associated with a group of the pillar structures **422** in a manner corresponding to (e.g., substantially the same as) that previously described in relation to operative association of the horizontal sense transistors **136** (FIGS. **1**A and **1**B) with the pillar groups **124** (FIG. **1**A) of the pillar structures **122** (FIGS. **1**A and **1**B).

(141) Referring next to FIG. **4**G, which is a simplified, longitudinal cross-sectional view of the portion of the microelectronic device structure **400** shown in FIG. **4**F following the processing stage previously described with reference to FIG. **4**F, isolation material **484** may be formed on or over exposed surfaces of the microelectronic device structure **400**. As shown in FIG. **4**G, the isolation material **484** may substantially fill the first openings **480** (FIG. **4**E) and the second openings **482** (FIG. **4**F). The isolation material **484** may be formed on surfaces of the SGD plug structures **426**, the gate dielectric structures **430**, the channel structures **434**, the backside dielectric structures **474**, and the dielectric cap structures **476**. An upper vertical boundary (e.g., an upper surface) of the isolation material **484** may be substantially planar, and a lower vertical boundary (e.g., a lower surface) of the isolation material **484** may be non-planar.

(142) The isolation material **484** may be formed of and include dielectric material, such as one or more of at least one dielectric oxide material (e.g., one or more of SiO.sub.x, AlO.sub.x, phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass), at least one dielectric nitride material (e.g., SiN.sub.y), and at least one low-K dielectric material (e.g., one or more of SiO.sub.xC.sub.y, SiO.sub.xN.sub.y, SiC.sub.xO.sub.yH.sub.z, and SiO.sub.xC.sub.zN.sub.y). The isolation material **484** may be substantially homogeneous, or the isolation material **484** may be heterogeneous. In some embodiments, the isolation material **484** is formed of and includes SiO.sub.x (e.g., SiO.sub.2).

(143) Referring next to FIG. **4**H, which is a simplified, longitudinal cross-sectional view of the portion of the microelectronic device structure **400** shown in FIG. **4**G following the processing stage previously described with reference to FIG. **4**G, an additional preliminary stack structure including preliminary selector tiers **437** each including additional sacrificial material **439** and additional insulative material **441** may be formed on or over the isolation material **484**; and then pillar openings **485** may be formed to vertically extend therethrough and to the horizontal sense transistors **436**. The pillar openings **485** may include first read pillar openings **485**A, second read pillar openings **485**B, and write pillar openings **485**C.

(144) The additional sacrificial material **439** of each of the preliminary selector tiers **437** of the additional preliminary stack structure may be formed of and include at least one material (e.g., at least one insulative material) that may be selectively removed relative to the additional insulative material **441** during subsequent processing (e.g., subsequent “replacement gate” or “gate last” processing) of the microelectronic device structure **400**, as described in further detail below. A material composition of the additional sacrificial material **439** is different than a material composition of the additional insulative material **441**. The material composition of the sacrificial material **407** and may be substantially the same as the material composition of the sacrificial material **407** of the preliminary tiers **409** of the preliminary stack structure vertically underlying the SGD plug structures **426**. The additional sacrificial material **439** may be selectively etchable relative to the additional insulative material **441** during common (e.g., collective, mutual) exposure to a first etchant, and the additional insulative material **441** may be selectively etchable to the additional sacrificial material **439** during common exposure to a second, different etchant. As a non-limiting example, the additional sacrificial material **439** may be formed of and include additional insulative material, such as one or more of at least one dielectric oxide material (e.g., one or more of SiO.sub.x, phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass, AlO.sub.x, HfO.sub.x, NbO.sub.x, TiO.sub.x, ZrO.sub.x, TaO.sub.x, and MgO.sub.x), at least one dielectric nitride material (e.g., SiN.sub.y), at least one dielectric oxynitride material (e.g., SiO.sub.xN.sub.y), and at least one dielectric carboxynitride material

(e.g., SiO.sub.xC.sub.zN.sub.y). In some embodiments, the additional sacrificial material **439** is formed of and includes dielectric nitride material, such as SiN y (e.g., Si.sub.3N.sub.4). The additional sacrificial material **439** of each of the preliminary selector tiers **437** may be substantially homogeneous, or the additional sacrificial material **439** of one or more (e.g., each) of the preliminary selector tiers **437** may be heterogeneous.
(145) The additional insulative material **441** of each of the preliminary selector tiers **437** of the additional preliminary stack structure may be formed of and include at least one dielectric material, such one or more of at least one dielectric oxide material (e.g., one or more of SiO.sub.x, phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass, AlO.sub.x, HfO.sub.x, NbO.sub.x, TiO.sub.x, ZrO.sub.x, TaO.sub.x, and MgO.sub.x), at least one dielectric nitride material (e.g., SiN.sub.y), at least one dielectric oxynitride material (e.g., SiO.sub.xN.sub.y), and at least one dielectric carboxynitride material (e.g., SiO.sub.xC.sub.zN.sub.y). In some embodiments, the additional insulative material **441** of each of the preliminary selector tiers **437** of the additional preliminary stack structure is formed of and includes dielectric oxide material, such as SiO.sub.x (e.g., SiO.sub.2). The additional insulative material **441** of each of the preliminary selector tiers **437** may be substantially homogeneous, or the additional insulative material **441** of one or more (e.g., each) of the preliminary selector tiers **437** may be heterogeneous.
(146) The additional preliminary stack structure may be formed to include any desired quantity of the preliminary selector tiers **437**. As shown in FIG. **411** in some embodiments, the additional preliminary stack structure is formed to include three (3) of the preliminary selector tiers **437**, a first preliminary selector tier **437**A, a second preliminary selector tier **437**B, and a third preliminary selector tier **437**C. In additional embodiments, the additional preliminary stack structure is formed to include a different quantity of the preliminary selector tiers **437**, such as two (2) of the preliminary selector tiers **437**, or greater than or equal to four (4) of the preliminary selector tiers **437**.
(147) Still referring to FIG. **4**H, the first read pillar openings **485**A and the second read pillar openings **485**B may be positioned within horizontal areas of the channel structures **434** (and, hence, the dielectric cap structures **476** and the backside dielectric structures **474**), and may vertical extend to or into the backside dielectric structures **474**. The write pillar openings **485**C may be positioned outside of the horizontal areas of the channel structures **434** (and, hence, the dielectric cap structures **476** and the backside dielectric structures **474**) but within horizontal areas of the SGD plug structures **426**, and may vertically extend to or into the SGD plug structures **426**. Within a horizontal area of an individual SGD plug structure **426**, one (1) of the first read pillar openings **485**A and one (1) of the second read pillar openings **485**B may be formed to vertically extend to and expose portions of one (1) of the backside dielectric structures **474**; and one (1) of the write pillar openings **485**C may be formed to vertically extend to and expose a portion of the SGD plug structures **426**. As shown in FIG. **4**H, an individual write pillar openings **485**C may be formed to vertically extend through the preliminary selector tiers **437** (including through the additional sacrificial material **439** and additional insulative material **441** thereof), the isolation material **484**, and an individual gate dielectric structures **430**, and into an individual SGD plug structure **426**. In addition, each of an individual first read pillar opening **485**A and an individual second read pillar openings **485**B may be formed to vertically extend through the preliminary selector tiers **437** (including through the additional sacrificial material **439** and additional insulative material **441** thereof), the isolation material **484**, and an individual dielectric cap structure **476**, and to or into an individual backside dielectric structure **474**.
(148) The first read pillar openings **485**A, the second read pillar openings **485**B, and the write pillar openings **485**C may be formed substantially simultaneously within one another. In some embodiments, the formation of the first read pillar openings **485**A, the second read pillar openings **485**B, and the write pillar openings **485**C utilizing multiple material removal processes. For

example, by way of first material removal process (e.g., a first anisotropic etching process), initial portions of the first read pillar openings **485**A and the second read pillar openings **485**B may be formed substantially simultaneously with one another and initial portions of the write pillar openings **485**C. The initial portions of the first read pillar openings **485**A and the second read pillar openings **485**B may vertically terminate at or within the dielectric cap structures **476**. The initial portions of the write pillar openings **485**C may vertically terminate at or within the SGD plug structures **426**. Next, by way of second material removal process (e.g., a second anisotropic etching process), the initial portions of the first read pillar openings **485**A and the second read pillar openings **485**B and the initial portions of the write pillar openings **485**C may be vertically extended substantially simultaneously with one another to form the first read pillar openings **485**A, the second read pillar openings **485**B, and the write pillar openings **485**C, respectively. The second material removal process may remove portions of the dielectric cap structures **476** exposed by the initial portions of the first read pillar openings **485**A and the second read pillar openings **485**B, and may also remove portions of the SGD plug structures **426** exposed by the initial portions of the write pillar openings **485**C.
(149) Referring next to FIG. **4**I, which is a simplified, longitudinal cross-sectional view of the portion of the microelectronic device structure **400** shown in FIG. **4**H following the processing stage previously described with reference to FIG. **4**H, gate dielectric material **488** may be formed inside and outside of the pillar openings **485** (FIG. **4**H), and then portions of the microelectronic device structure **400** vertically underlying and within horizontal areas of the pillar openings **485** (FIG. **4**H) may be removed to vertically extend the pillar openings **485** (FIG. **4**H) and form extended pillar openings **486**. The extended pillar openings **486** may include first extended read pillar openings **486**A, second extended read pillar openings **486**B, and extended write pillar openings **486**C.
(150) As shown in FIG. **4**I, the gate dielectric material **488** may be formed on and may substantially cover side surfaces of the preliminary selector tiers **437** (including side surfaces of the additional sacrificial material **439** and additional insulative material **441** thereof) partially defining the pillar openings **485** (FIG. **4**H). The gate dielectric material **488** may also be formed on and may substantially cover an upper surface of an uppermost one of the preliminary selector tiers **437** outside of the horizontal boundaries of the pillar openings **485** (FIG. **4**H). For portions of gate dielectric material **488** partially defining the first extended read pillar openings **486**A and the second extended read pillar openings **486**B, the gate dielectric material **488** may also be formed on and may substantially cover side surfaces of the dielectric cap structures **476**, and may vertically terminate at or within the backside dielectric structures **474**. Vertical dimensions of the portions of the gate dielectric material **488** partially defining the first extended read pillar openings **486**A and the second extended read pillar openings **486**B may be less than the vertical dimensions of the first extended read pillar openings **486**A and the second extended read pillar openings **486**B. Furthermore, for additional portions of the gate dielectric material **488** partially defining the extended write pillar openings **486**C, the gate dielectric material **488** may vertically terminate at or within the SGD plug structures **426**. Vertical dimensions of the additional portions of the gate dielectric material **488** partially defining the extended write pillar openings **486**C may be less than the vertical dimensions of the extended write pillar openings **486**C.
(151) The gate dielectric material **488** may be formed of and include dielectric material, such as one or more of at least one dielectric oxide material (e.g., one or more of SiO.sub.x, AlO.sub.x, phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass), at least one dielectric nitride material (e.g., SiN y), and at least one low-K dielectric material (e.g., one or more of SiO.sub.xC.sub.y, SiO.sub.xN.sub.y, SiC.sub.xO.sub.yH.sub.z, and SiO.sub.xC.sub.zN.sub.y). The gate dielectric material **488** may be substantially homogeneous, or the gate dielectric material **488** may be heterogeneous. In some embodiments, the gate dielectric material **488** is formed of and includes SiO.sub.x (e.g., SiO.sub.2).

(152) Still referring to FIG. **4**I, the first extended read pillar openings **486**A and the second extended read pillar openings **486**B may have horizontal boundaries defined by side surfaces of the gate dielectric material **488**, the backside dielectric structures **474**, and the channel structures **434**; and may have lower vertical boundaries defined by upper surfaces of the gate dielectric structures **430**. Horizontal dimensions of portions of the first extended read pillar openings **486**A and the second extended read pillar openings **486**B at vertical positions of the backside dielectric structures **474** and lower portions of the dielectric cap structures **476** may be relatively smaller (e.g., horizontally narrower) than horizontal dimensions of additional portions of the first extended read pillar openings **486**A and the second extended read pillar openings **486**B at vertical positions of the channel structures **434** and the preliminary selector tiers **437**.
(153) Still referring to FIG. **4**I, the extended write pillar openings **486**C may have horizontal boundaries defined by side surfaces of the gate dielectric material **488** and the SGD plug structures **426**; and may have lower vertical boundaries defined by surfaces of the SGD plug structures **426**. Horizontal dimensions of portions of the extended write pillar openings **486**C at relatively higher vertical positions within the SGD plug structures **426** may be relatively smaller (e.g., horizontally narrower) than horizontal dimensions of additional portions of the first extended read pillar openings **486**A and the second extended read pillar openings **486**B at relatively lower vertical positions within the SGD plug structures **426** and at vertical positions of isolation material **484** and the preliminary selector tiers **437**.
(154) To form the gate dielectric material **488** and the extended pillar openings **486**, the gate dielectric material **488** may be formed inside and outside of the pillar openings **485** (FIG. **4**H) and may partially (e.g., less than completely) fill the pillar openings **485** (FIG. **4**H). Thereafter, a sacrificial liner material, such as a semiconductor liner material (e.g., polycrystalline silicon liner) may be formed on or over the gate dielectric material **488**, and may partially (e.g., less than completely) fill remaining portions of the pillar openings **485** (FIG. **4**H). A relative thickness of the sacrificial liner material formed within the pillar openings **485** (FIG. **4**H) is represented by way of the dashed line **490** within the extended write pillar opening **486**C shown in FIG. **4**I. Following the formation of sacrificial liner material, portions of the semiconductor liner material and the gate dielectric material **488** proximate lower vertical boundaries (e.g., bottoms) of the pillar openings **485** (FIG. **4**H) may be removed, such as by way of a so-called “punch through” etch. Within horizontal areas of the first read pillar openings **485**A (FIG. **4**H) and the second read pillar openings **485**B (FIG. **4**H), the material removal process may expose positions of the channel structures **434**. Within horizontal areas of the write pillar openings **485**C (FIG. **4**H), material removal process may expose positions of the SGD plug structures **426**. Thereafter, at least one additional material removal process (e.g., at least one etching process, such as at least one wet etching process) may be performed to remove the semiconductor liner material, as well as exposed portions of the channel structures **434** and expose positions of the SGD plug structures **426**. The additional material removal process may vertically recess and horizontally recess portions of the channel structures **434**, and may complete the formation of the first extended read pillar openings **486**A and the second extended read pillar openings **486**B. The additional material removal process may also vertically recess and horizontally recess the portions of the SGD plug structures **426**, and may complete the formation of the extended write pillar openings **486**C.
(155) Referring next to FIG. **4**J, which is a simplified, longitudinal cross-sectional view of the portion of the microelectronic device structure **400** shown in FIG. **4**J following the processing stage previously described with reference to FIG. **4**J, additional channel material **492**, additional dielectric fill material **494**, and plug material **496** may be formed inside and outside of the extended pillar openings **486** (FIG. **4**I) (including the first extended read pillar openings **486**A (FIG. **4**I), the second extended read pillar openings **486**B (FIG. **4**I), and the extended write pillar openings **486**C (FIG. **4**I)); and then portions of the gate dielectric material **488**, the additional channel material **492**, the additional dielectric fill material **494**, and the plug material **496** outside of the boundaries

of the extended pillar openings **486** (FIG. **4**I) may be removed (e.g., by way of a CMP process) to form write select pillar structures **444**, first read select pillar structures **446**, and second read select pillar structures **448**. The write select pillar structures **444**, the first read select pillar structures **446**, and the second read select pillar structures **448** may respectively be formed to have configurations corresponding to (e.g., substantially the same as) the configurations of the write select pillar structures **144**, the first read select pillar structures **146**, and the second read select pillar structures **148** previously described with reference to FIGS. **1**A and **1**B.

(156) As shown in FIG. **4**J, the additional channel material **492** may be formed to substantially fill portions of the extended write pillar openings **486**C (FIG. **4**I) at the vertical position of the SGD plug structures **426**. Accordingly, the additional channel material **492** of the write select pillar structures **444** may be formed to contact (e.g., physically contact, electrically contact) the SGD plug structures **426** of the horizontal sense transistors **436**. The additional channel material **492** may also substantially fill additional portions of the extended write pillar openings **486**C (FIG. **4**I) at the vertical positions of the gate dielectric structures **430**. In addition, additional channel material **492** may partially fill further portions of the extended write pillar openings **486**C (FIG. **4**I) vertically overlying the additional portions of the extended write pillar openings **486**C (FIG. **4**I), such as portions extending from the gate dielectric structures **430** to or beyond an upper boundary of an uppermost one of the preliminary selector tiers **437**.

(157) As also shown in FIG. **4**J, the additional channel material **492** may be formed to substantially fill portions of the first extended read pillar openings **486**A (FIG. **4**I) and the second extended read pillar openings **486**B (FIG. **4**I) at the vertical position of the channel structures **434**. Accordingly, the additional channel material **492** of the first read select pillar structures **446** and the second read select pillar structures **448** may be formed to contact (e.g., physically contact, electrically contact) the channel structures **434** of the horizontal sense transistors **436**. The additional channel material **492** may also substantially fill additional portions of the first extended read pillar openings **486**A (FIG. **4**I) and the second extended read pillar openings **486**B (FIG. **4**I) at the vertical positions of the backside dielectric structures **474** and lower portions of the dielectric cap structures **476**. In addition, additional channel material **492** may partially fill further portions of the first extended read pillar openings **486**A (FIG. **4**I) and the second extended read pillar openings **486**B (FIG. **4**I) vertically overlying the additional portions of the first extended read pillar openings **486**A (FIG. **4**I) and the second extended read pillar openings **486**B (FIG. **4**I), such as portions extending from the lower portions of the dielectric cap structures **476** to or beyond an upper boundary of an uppermost one of the preliminary selector tiers **437**.

(158) The additional dielectric fill material **494** may partially fill portions of the extended pillar openings **486** (FIG. **4**I) not occupied by the additional channel material **492**. For each of the write select pillar structures **444**, the first read select pillar structures **446**, and the second read select pillar structures **448**, surfaces of the additional channel material **492** may define horizontal boundaries and lower vertical boundaries of the additional dielectric fill material **494**; and an upper vertical boundary (e.g., an upper surface) of the additional dielectric fill material **494** may be formed to be vertically recessed (e.g., to vertically underlie) relative to an upper vertical boundary of the additional channel material **492**. The additional dielectric fill material **494** may be formed on the additional channel material **492**, and may be formed of and include dielectric material (e.g., one or more of dielectric oxide material, dielectric nitride material, and an air gap).

(159) The plug material **496** may fill upper portions of the extended pillar openings **486** (FIG. **4**I) not occupied by the additional channel material **492** and the additional dielectric fill material **494**. For each of the write select pillar structures **444**, the first read select pillar structures **446**, and the second read select pillar structures **448**, side surfaces of the additional channel material **492** may define horizontal boundaries of the plug material **496**, and upper surfaces of the additional dielectric fill material **494** may define lower vertical boundaries of the plug material **496**. An upper vertical boundary (e.g., an upper surface) of the plug material **496** may be formed to be

substantially coplanar with to the upper vertical boundary (e.g., the upper surface) of the additional channel material **492**. The plug material **496** may be formed on the additional dielectric fill material **494** and the additional channel material **492**. The plug material **496** may be formed of and include conductive material, such as semiconductor material (e.g., polycrystalline silicon) doped with doped with at least one conductivity-enhancing dopant (e.g., at least one N-type dopant, such as one or more of P, As, Sb, and Bi; at least one P-type dopant, such as one or more of B, Al, and Ga). In some embodiments, the plug material **496** is formed of and includes N-type polycrystalline silicon (e.g., polycrystalline silicon doped with at least one N-type dopant, such as polycrystalline silicon doped with phosphorous (P)).
(160) Referring next to FIG. **4**K, which is a simplified, longitudinal cross-sectional view of the portion of the microelectronic device structure **400** shown in FIG. **4**J following the processing stage previously described with reference to FIG. **4**J, at least one third masking structure **497** including openings vertically extending therethrough may be formed on or over an uppermost one of the preliminary selector tiers **437** (e.g., on or over the third preliminary selector tier **437**C), and then one or more conductivity-enhancing species may be provided (e.g., implanted) into one or more of the preliminary selector tiers **437** at horizontal positions of the openings in the third masking structure **497**. The doping process may form some doped sections **442** (e.g., first doped sections **442**A) and some further doped sections **443** at desired vertical positions within vertical boundaries of the preliminary selector tiers **437**. As a non-limiting example, first doped sections **442**A corresponding to the first doped sections **142**A previously described with reference to FIG. **1**A may be formed at horizontal positions and vertical locations corresponding to those previously described with reference to FIG. **1**A. As another non-limiting example, further doped sections **443** may be formed to vertically overlie and at least partially horizontally overlap the first doped sections **442**A.
(161) In some embodiments, the third masking structure **497** is a photoresist mask formed of and including photoresist material, such as positive tone photoresist material or negative tone photoresist material. The third masking structure **497** may, for example, be compatible with 13.7 nm, 157 nm, 193 nm, 248 nm, or 365 nm wavelength systems; with 193 nm wavelength immersion systems; and/or with electron beam lithographic systems. In addition, the openings in the third masking structure **497** may be formed using conventional processes (e.g., conventional photolithographic patterning and development processes) and conventional equipment, which are not described in detail herein.
(162) To form the first doped sections **442**A, at least one conductivity-enhancing species (e.g., at least one P-type dopant) may be implanted into portions of the first preliminary selector tier **437**A within horizontal areas of the openings in the third masking structure **497** and into portions of the write select pillar structures **444** at the vertical position of the first preliminary selector tier **437**A. In some embodiments, the conductivity-enhancing species includes at least one P-type dopant (e.g., one or more of B, Al, and Ga). The conductivity-enhancing species (e.g., P-type dopant) may be implanted using conventional implantation processes (e.g., a PLAD implantation process, a BL implantation process) and equipment, which are not described in detail herein.
(163) To form the further doped sections **443**, at least one additional conductivity-enhancing species (e.g., at least one N-type dopant) different than that employed in the first doped sections **442**A may be implanted into portions of the second preliminary selector tier **437**B within the horizontal areas of the openings in the third masking structure **497** and into portions of the write select pillar structures **444** at the vertical position of the second preliminary selector tier **437**B. In some embodiments, the additional conductivity-enhancing species includes at least one N-type dopant (e.g., one or more of P, As, Sb, and Bi). The further doped sections **443** may be formed after the formation of the first doped sections **442**A. The additional conductivity-enhancing species (e.g., N-type dopant) may be implanted using conventional implantation processes (e.g., a PLAD implantation process, a BL implantation process) and equipment, which are not described in detail herein.

(164) Referring next to FIG. **4**L, which is a simplified, longitudinal cross-sectional view of the portion of the microelectronic device structure **400** shown in FIG. **4**K following the processing stage previously described with reference to FIG. **4**K, one or more additional conductivity-enhancing species may be provided (e.g., implanted) into one or more of the preliminary selector tiers **437**. The doping process may at least form some more doped sections **442** (e.g., second doped sections **442**B, third doped sections **442**C) at desired vertical positions within vertical boundaries of the preliminary selector tiers **437**. As a non-limiting example, second doped sections **442**B and third doped sections **442**C respectively corresponding to the second doped sections **142**B and the third doped sections **142**C previously described with reference to FIG. **1**A may be formed at horizontal positions and vertical locations corresponding to those previously described with reference to FIG. **1**A.
(165) To form the second doped sections **442**B, at least one conductivity-enhancing species (e.g., at least one P-type dopant) may be implanted into portions of the second preliminary selector tier **437**B and at least into portions of the first read select pillar structures **446** and the second read select pillar structures **448** at the vertical position of the second preliminary selector tier **437**B. In some embodiments, the conductivity-enhancing species includes at least one P-type dopant (e.g., one or more of B, Al, and Ga). Optionally, the conductivity-enhancing species may also implanted into portions of the write select pillar structures **444** at the vertical position of the second preliminary selector tier **437**B, such that the portions of the write select pillar structures **444** are doped with at least one conductivity-enhancing species (e.g., at least one N-type dopant) as a result of the processing stage of FIG. **4**K and are also doped with at least one different conductively enhancing species (e.g., at least one P-type dopant) as a result of the processing stage of FIG. **4**K. As a result, the write selector transistors subsequently formed at the portions of the write select pillar structures **444** at the vertical position of the second preliminary selector tier **437**B may have different V.sub.t (e.g., negative V.sub.t) characteristics than V.sub.t (e.g., positive V.sub.t) characteristics of read selector transistors subsequently formed at the portions of the read select pillar structures **446** at the vertical position of the second preliminary selector tier **437**B. The conductivity-enhancing species (e.g., P-type dopant) may be implanted during that processing stage of FIG. **4**L using conventional implantation processes (e.g., a PLAD implantation process, a BL implantation process) and equipment, which are not described in detail herein.
(166) To form the third doped sections **442**C, at least one conductivity-enhancing species (e.g., at least one P-type dopant) may be implanted into portions of the third preliminary selector tier **437**C and into portions of the write select pillar structures **444**, the first read select pillar structures **446**, and the second read select pillar structures **448** at the vertical position of the third preliminary selector tier **437**C. In some embodiments, the conductivity-enhancing species includes at least one P-type dopant (e.g., one or more of B, Al, and Ga). The conductivity-enhancing species (e.g., P-type dopant) may be implanted using conventional implantation processes (e.g., a PLAD implantation process, a BL implantation process) and equipment, which are not described in detail herein.
(167) Referring next to FIG. **4**M, which is a simplified, longitudinal cross-sectional view of the portion of the microelectronic device structure **400** shown in FIG. **4**M following the processing stage previously described with reference to FIG. **4**M, the preliminary tiers **409** (FIG. **4**K) and the preliminary selector tiers **437** (FIG. **4**K) may be subjected to so-called “replacement gate” or “gate last” processing to form tiers **410** from the preliminary tiers **409** (FIG. **4**K) and selector tiers **428** from the preliminary selector tiers **437** (FIG. **4**K). The tiers **410** may each include at least one conductive structure **406** vertically neighboring at least one insulative structure **408**; and the selector tiers **438** may each include at least one additional conductive structure **440** vertically neighboring a remaining portion of the additional insulative material **441**. Following the replacement gate process, read source line structures **452** may be formed vertically over and in contact (e.g., physical contact, electrical contact) with the first read select pillar structures **446**; and

local strap structures **454** may be formed vertically over and in contact (e.g., physical contact, electrical contact) with the write select pillar structures **444** and the second read select pillar structures **448**. Thereafter, digit line contact structures **460** may be formed vertically over and in contact (e.g., physical contact, electrical contact) with the local strap structures **454**; and digit line structures **458** may be formed vertically over and in contact with the digit line contact structures **460**. The tiers **410**, the selector tiers **438**, the read source line structures **452**, the local strap structures **454**, the digit line contact structures **460**, and the digit line structures **458** may respectively be formed to have configurations corresponding to (e.g., substantially the same as) the configurations of the tiers **110**, the selector tiers **138**, the read source line structures **152**, the local strap structures **154**, the digit line contact structures **160**, and the digit line structures **158** previously described with reference to FIGS. **1**A and **1**B.

(168) Referring to back to FIG. **4**K, the replacement gate process effectuated at the processing stage of FIG. **4**M may include forming slots (e.g., slits, openings) vertically extending through the preliminary selector tiers **437** and the preliminary tiers **409**, and then treating the microelectronic device structure **400** with at least one wet etchant formulated to selectively remove portions of the sacrificial material **407** of the preliminary tiers **409** and the additional sacrificial material **439** of the preliminary selector tiers **437** through the slots. The wet etchant may be selected to remove the portions of the sacrificial material **407** and the additional sacrificial material **439** without substantially removing portions of the insulative material **405** of the preliminary tiers **409** and additional insulative material **441** of the preliminary selector tiers **437**. In some embodiments wherein the sacrificial material **407** and the additional sacrificial material **439** comprise dielectric nitride material (e.g., SiN.sub.y, such as Si.sub.3N.sub.4) and the insulative material **405** and the additional insulative material **441** comprise dielectric oxide material (e.g., SiO.sub.x, such as SiO.sub.2), the sacrificial material **407** and the additional sacrificial material **439** are selectively removed using a wet etchant comprising H.sub.3PO.sub.4. Following the selective removal of the portions of the sacrificial material **407** and the additional sacrificial material **439**, the resulting recesses may be filled with conductive material to form the conductive structures **406** (FIG. **4**M) and the additional conductive structures **440** (FIG. **4**M). The conductive structures **406** (FIG. **4**M) and the additional conductive structures **440** (FIG. **4**M) may be formed substantially simultaneously with one another through the replacement gate process.

(169) Referring again to FIG. **4**M, the replacement gate process may effectuate the formation of select transistors **423** within vertical boundaries of the tiers **410** of the microelectronic device structure **400**, as well as the formation of write selector transistors **431** and read selector transistors **435** within vertical boundaries of the selector tiers **438** of the microelectronic device structure **400**. The select transistors **423**, the write selector transistors **431**, and read selector transistors **435** may respectively have configurations corresponding to (e.g., substantially the same as) the configurations of select transistors **123**, the write selector transistors **131**, and read selector transistors **135** previously described with reference to FIGS. **1**A and **1**B. In addition, the replacement gate process may also effectuate the formation of strings of memory cells operatively associated with the pillar structures **422**, wherein the strings of memory cells have configurations corresponding to (e.g., substantially the same as) the configurations of the strings of memory cells **121** previously described with reference to FIGS. **1**A and **1**B.

(170) Following the replacement gate process, the read source line structures **452**, the local strap structures **454**, the digit line contact structures **460**, the digit line structures **458**, and different portions of at least one isolation material **462** may be individually formed using conventional processes (e.g., convention material deposition processes, conventional photolithographic patterning processes, conventional material removal processes) and conventional processing equipment, which are not described in detail herein.

(171) Thus, in accordance with embodiments of the disclosure, a method of forming a microelectronic device comprises forming pillar structures to vertically extend through a stack

structure comprising tiers each including sacrificial material vertically adjacent insulative material. P-type dopant is implanted into different horizontal portions of some of the tiers and into portions of the pillar structures at vertical positions of the some of the tiers. Sense transistors are formed vertically over the stack structure. The sense transistors are coupled to different groups of the pillar structures than one another. An additional stack structure is formed over the sense transistors. The additional stack structure comprises additional tiers each including additional sacrificial material vertically adjacent additional insulative material. Additional pillar structures are formed to vertically extend through the additional stack structure and to the sense transistors. Additional P-type dopant is implanted into different horizontal portions of two or more of the additional tiers and into portions of the additional pillar structures at vertical elevations of the two or more of the additional tiers. The sacrificial material and the additional sacrificial material are replaced with conductive material. Conductive line structures are formed over and in electrical communication with the additional pillar structures.

(172) Microelectronic devices structures (e.g., the microelectronic device structures **100**, **200**, **300**, **400**) and microelectronic devices in accordance with embodiments of the disclosure may be used in embodiments of electronic systems of the disclosure. For example, FIG. **5** is a schematic block diagram of an illustrative electronic system **500** according to embodiments of disclosure. The electronic system **500** may comprise, for example, a computer or computer hardware component, a server or other networking hardware component, a cellular telephone, a digital camera, a personal digital assistant (PDA), portable media (e.g., music) player, a Wi-Fi or cellular-enabled tablet such as, for example, one or more of an iPad® or SURFACE® tablet, an electronic book, and a navigation device. The electronic system **500** includes at least one memory device **502**. The memory device **502** may comprise, for example, one or more of a microelectronic device structure (e.g., one or more of the microelectronic device structures **100**, **200**, **300**, **400**) and a microelectronic device previously described herein. The electronic system **500** may further include at least one electronic signal processor device **504** (often referred to as a “microprocessor”). The electronic signal processor device **504** may, optionally, include one or more of a microelectronic device structure (e.g., one or more of the microelectronic device structures **100**, **200**, **300**, **400**) and a microelectronic device previously described herein. While the memory device **502** and the electronic signal processor device **504** are depicted as two (2) separate devices in FIG. **5**, in additional embodiments, a single (e.g., only one) memory/processor device having the functionalities of the memory device **502** and the electronic signal processor device **504** is included in the electronic system **500**. In such embodiments, the memory/processor device may include one or more of a microelectronic device structure (e.g., one or more of the microelectronic device structures **100**, **200**, **300**, **400**) and a microelectronic device previously described herein. The electronic system **500** may further include one or more input devices **506** for inputting information into the electronic system **500** by a user, such as, for example, a mouse or other pointing device, a keyboard, a touchpad, a button, or a control panel. The electronic system **500** may further include one or more output devices **508** for outputting information (e.g., visual or audio output) to a user such as, for example, one or more of a monitor, a display, a printer, an audio output jack, and a speaker. In some embodiments, the input device **506** and the output device **508** may comprise a single touchscreen device that can be used both to input information to the electronic system **500** and to output visual information to a user. The input device **506** and the output device **508** may communicate electrically with one or more of the memory device **502** and the electronic signal processor device **504**.

(173) Thus, an electronic system in accordance with embodiments of the disclosure comprises an input device, an output device, a processor device operably connected to the input device and the output device, and a memory device operably connected to the processor device. The memory device comprises pillar structures vertically extending through blocks of a stack structure comprising conductive material vertically alternating with insulative material; conductive plug

structures within horizontal areas of the blocks of the stack structure and coupled to different groups of the pillar structures than one another, each of the different groups of the pillar structures including multiple of the pillar structures positioned within different sub-blocks of one of the blocks than one another; horizontal sense transistors gated by the conductive plug structures; read source lines vertically overlying the horizontal sense transistors; digit lines vertical overlying the read source lines; first vertical read selector transistors vertically interposed between and in electrical communication with the read source lines and source regions of the horizontal sense transistors; second vertical read selector transistors vertically interposed between and in electrical communication with the digit lines structures and drain regions of the horizontal sense transistors; and vertical write selector transistors vertically interposed between and in electrical communication with the digit lines structures and the conductive plug structures.
(174) The structures, devices, methods, and systems of the disclosure advantageously facilitate one or more of improved microelectronic device performance, reduced costs (e.g., manufacturing costs, material costs), increased miniaturization of components, and greater packaging density as compared to conventional structures, conventional devices, conventional methods, and conventional systems. The structure, devices, methods, and systems of the disclosure may also improve scalability, efficiency, and simplicity as compared to conventional structures, conventional devices, conventional methods, and conventional systems.
(175) While the disclosure is susceptible to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and have been described in detail herein. However, the disclosure is not limited to the particular forms disclosed. Rather, the disclosure is to cover all modifications, equivalents, and alternatives falling within the scope of the following appended claims and their legal equivalent. For example, elements and features disclosed in relation to one embodiment may be combined with elements and features disclosed in relation to other embodiments of the disclosure.

Claims

1. A microelectronic device, comprising: a stack structure comprising a vertically alternating sequence of conductive material and insulative material, the stack structure divided into blocks separated by dielectric slot structures, the blocks individually including sub-blocks horizontally extending in parallel with one another; pillar structures comprising semiconductor material vertically extending through one of the blocks of the stack structure, each pillar structure of a group of the pillar structures horizontally positioned within a different one of the sub-blocks of the one of the blocks than each other pillar structure of the group of the pillar structures; a conductive plug structure coupled to and horizontally extending across and between multiple of the pillar structures of the group of the pillar structures; a sense transistor gated by the conductive plug structure; and selector transistors coupling the sense transistor to a read source line structure and a digit line structure.
2. The microelectronic device of claim 1, wherein the stack structure comprises tiers each including the conductive material and the insulative material vertically adjacent the conductive material, the tiers grouped into tier sections comprising: an access line tier section comprising a first group of the tiers; stacked drain side select gate (SGD) tier section overlying the access line tier section and comprising a second group of the tiers; and a sense node tier section overlying the SGD tier section and comprising a third group of the tiers.
3. The microelectronic device of claim 2, wherein the conductive plug structure and portions of each of the second group of the tiers and the third group of the tiers within a horizontal area of the one of the blocks of the stack structure occupied by the group of the pillar structures define a sense node of the one of the blocks.
4. The microelectronic device of claim 2, wherein: the second group of the tiers of the SGD tier

section comprises: a program-inhibit SGD tier overlying the access line tier section; multiple SGD tiers overlying the program-inhibit SGD tier; and a read-amplification SGD tier overlying a second SGD tier; and the third group of the tiers of the sense node tier section comprises multiple select gate programming (SGP) tiers overlying the read-amplification SGD tier.
5. The microelectronic device of claim 4, wherein: the multiple SGD tiers comprise: a first SGD bar tier overlying the program-inhibit SGD tier; a second SGD bar tier overlying the first SGD bar tier; a first SGD tier overlying the second SGD bar tier; and a second SGD tier overlying the first SGD tier; and the multiple SGP tiers comprise: a first SGP bar tier overlying the read-amplification SGD tier; a second SGP bar tier overlying the first SGP bar tier; a first SGP tier overlying the second SGP bar tier; and a second SGP tier overlying the first SGP tier.
6. The microelectronic device of claim 5, wherein each of the first SGD bar tier, the second SGD bar tier, the first SGD tier, and the second SGD tier comprises a different horizontal arrangement of higher V.sub.t transistors and lower V.sub.t transistors within vertical boundaries thereof than each other of the first SGD bar tier, the second SGD bar tier, the first SGD tier, and the second SGD tier.
7. The microelectronic device of claim 5, wherein each of the first SGP bar tier, the second SGP bar tier, the first SGP tier, and the second SGP tier comprises a different horizontal arrangement of higher V.sub.t transistors and lower V.sub.t transistors within vertical boundaries thereof than each other of the first SGP bar tier, the second SGP bar tier, the first SGP tier, and the second SGP tier.
8. The microelectronic device of claim 4, wherein intersections of the pillar structures and the conductive material of at least some of the tiers within of the second group of the tiers and the third group of the tiers define select transistors within the blocks of the stack structure, some of the select transistors having a relatively higher threshold voltage level than some other of the select transistors.
9. The microelectronic device of claim 4, wherein the third group of the tiers of the sense node tier section further comprises a gate-induced drain-leakage (GIDL) generation tier overlying the multiple SGP tiers.
10. The microelectronic device of claim 1, wherein the group of the pillar structures comprises four of the pillar structures.
11. The microelectronic device of claim 1, wherein: the sub-blocks of the blocks horizontally extend in parallel with the dielectric slot structures in a first direction; and a horizontal orientation the conductive plug structure is acutely angled relative to the first direction.
12. The microelectronic device of claim 1, wherein: conductive plug structure horizontally extends in parallel with the dielectric slot structures in a first direction; and the sub-blocks of the blocks horizontally extend in parallel in a second direction acutely angled relative to the first direction.
13. The microelectronic device of claim 1, wherein the sense transistor comprises a horizontal sense transistor comprising: the conductive plug structure; a channel structure vertically overlying and only partially horizontally extending across the conductive plug structure; and a gate dielectric structure vertically interposed between the conductive plug structure and the channel structure.
14. The microelectronic device of claim 13, wherein the selector transistors comprise vertical selector transistors, each of the vertical selector transistors comprising: a semiconductive material comprising channel region vertically interposed between a source region and a drain region; a gate electrode horizontally neighboring the semiconductive material and within vertical boundaries of the channel region; and a gate dielectric material horizontally interposed between the semiconductive material and the gate electrode.
15. The microelectronic device of claim 1, the selector transistors comprise: a write selector transistor vertically interposed between and coupled to the digit line structure and the conductive plug structure; a first read selector transistor vertically interposed between and coupled to the read source line structure and a source side of a channel structure of the sense transistor; and a second read selector transistor vertically interposed between and coupled to the digit line structure and a drain side of the channel structure of the sense transistor.

16. The microelectronic device of claim 15, further comprising: a local strap structure coupled to each of the digit line structure, the write selector transistor, and the second read selector transistor; and a digit line contact vertically interposed between and coupled to the local strap structure and the digit line structure.
17. The microelectronic device of claim 1, wherein: the read source line structure vertically overlies the sense transistor; the digit line structure vertically overlies the read source line structure; and the selector transistors are vertically interposed between the read source line structure and the sense transistor.
18. The microelectronic device of claim 17, wherein the selector transistors comprise: a write selector transistor vertically interposed between and coupled to the digit line structure and the conductive plug structure; a read selector transistor vertically interposed between and coupled to one of: the read source line structure and a source side of a channel structure of the sense transistor; and the digit line structure and a drain side of the channel structure of the sense transistor.
19. The microelectronic device of claim 18, further comprising an additional read selector transistor having different V.sub.t characteristics than the read selector transistor, the additional read selector transistor vertically interposed between and coupled to an other of: the read source line structure and the source side of the channel structure of the sense transistor; and the digit line structure and the drain side of the channel structure of the sense transistor.
20. The microelectronic device of claim 18, further comprising a conductive contact structure vertically interposed between and coupled to an other of: the read source line structure and the source side of the channel structure of the sense transistor; and the digit line structure and the drain side of the channel structure of the sense transistor.
21. A method of forming a microelectronic device, comprising: forming pillar structures to vertically extend through a stack structure comprising tiers each including sacrificial material vertically adjacent insulative material; implanting P-type dopant into different horizontal portions of some of the tiers and into portions of the pillar structures at vertical positions of the some of the tiers; forming sense transistors vertically over the stack structure, the sense transistors coupled to different groups of the pillar structures than one another; forming an additional stack structure over the sense transistors, the additional stack structure comprising additional tiers each including additional sacrificial material vertically adjacent additional insulative material; forming additional pillar structures to vertically extend through the additional stack structure and to the sense transistors; implanting additional P-type dopant into different horizontal portions of two or more of the additional tiers and into portions of the additional pillar structures at vertical elevations of the two or more of the additional tiers; replacing the sacrificial material and the additional sacrificial material with conductive material; and forming conductive line structures over and in electrical communication with the additional pillar structures.
22. The method of claim 21, wherein forming pillar structures comprises forming the pillar structures to individually comprise: cell film material on surfaces of the stack structure; semiconductive channel material inwardly horizontally adjacent the cell film material; and dielectric fill material inwardly horizontally adjacent the semiconductive channel material.
23. The method of claim 21, further comprising selecting the P-type dopant to comprise boron.
24. The method of claim 21, wherein implanting the P-type dopant comprises forming a different pattern of regions doped with the P-type dopant at a vertical position of at least one tier of the some of the tiers than at a vertical position of at least one other tier of the some of the tiers.
25. The method of claim 21, wherein forming sense transistors vertically over the stack structure comprises: forming conductive plug material over the stack structure and the pillar structures; forming gate dielectric material over the conductive plug material; forming channel material over the gate dielectric material; forming back-side dielectric material over the channel material; forming dielectric cap material over the back-side dielectric material; and removing portions of the dielectric cap material, the back-side dielectric material, the channel material, the gate dielectric

material, and the conductive plug material to respectively form dielectric cap structures, back-side dielectric structures, channel structures, gate dielectric structures, and conductive plug structures.
26. The method of claim 25, further comprising: forming each of the conductive plug structures to be coupled to and horizontally extend across and between a different group of multiple of the pillar structures than each other of the conductive plug structures; and forming the channel structures to be substantially confined within horizontal areas of the conductive plug structures and to have additional horizontal areas smaller than the horizontal areas of the conductive plug structures.
27. The method of claim 26, wherein forming additional pillar structures to vertically extend through the additional stack structure and to the sense transistors comprises: forming read select pillar structures vertically extending through the additional stack structure, the dielectric cap structures, and the back-side dielectric material and into the channel structures; and forming write select pillar structures vertically extending through the additional stack structure and into the conductive plug structures.
28. The method of claim 21, wherein replacing the sacrificial material and the additional sacrificial material with conductive material comprises: forming slots vertically extending through the additional stack structure and the stack structure; simultaneously removing the sacrificial material of the tiers of the stack structure and the additional sacrificial material of the additional tiers of the additional stack structure by way of the slots to form voids in the tiers of the stack structure and additional voids in the additional tiers of the additional stack structure; and simultaneously filling the voids in the tiers of the stack structure and the additional voids in the additional tiers of the additional stack structure with the conductive material.
29. The method of claim 21, wherein forming conductive line structures over and in electrical communication with the additional pillar structures comprises: forming read source line structures over and in electrical communication with some of the additional pillar structures coupled to channel structures of the sense transistors; and forming digit line structures over and in electrical communication with: some other of the additional pillar structures coupled to the channel structures of the sense transistors, and yet some other of the additional pillar structures coupled to gate electrodes of the sense transistors.
30. A memory device, comprising: a stack structure divided into blocks separated by dielectric slot structures, each of the blocks comprising: an access line section comprising tiers including access line structures; a select gate section overlying the access line section and comprising additional tiers including drain side select gate (SGD) structures; and a sense node section overlying the select gate section and comprising further tiers including select gate programming (SGP) structures; pillar groups within horizontal areas of the blocks and individually comprising multiple pillar structures vertically extending completely through one of the blocks, each pillar structure of the multiple pillar structures horizontally positioned within a different sub-block of the one of the blocks than each other pillar structure of the multiple pillar structures; horizontal sense transistors vertically overlying and coupled to the pillar groups; vertical read selector transistors vertically overlying and coupled to the horizontal sense transistors; vertical write selector transistors vertically overlying and coupled to the horizontal sense transistors, the vertical write selector transistors horizontally offset from the vertical read selector transistors; and conductive line structures vertically overlying and coupled to the vertical read selector transistors and the vertical write selector transistors.
31. The memory device of claim 30, wherein the SGD structures of each of the additional tiers of at least one of the blocks continuously horizontally extend from and between at least two of the dielectric slot structures horizontally neighboring the at least one of the blocks.
32. The memory device of claim 30, further comprising, horizontal areas of the blocks: select gate electrodes vertically overlying the horizontal sense transistors; read select pillar structures vertically extending through select gate electrodes and coupled to channel structures of the horizontal sense transistors, intersections of the read select pillar structures and the select gate electrodes defining the vertical read selector transistors; and write select pillar structures vertically

extending through select gate electrodes and coupled to gate electrodes of the horizontal sense transistors, intersections of the write select pillar structures and the select gate electrodes defining the vertical write selector transistors.

33. The memory device of claim 32, wherein each of the horizontal sense transistors is individually coupled to two of the read select pillar structures and one of the write select pillar structures.

34. The memory device of claim 30, wherein the conductive line structures comprise read source line structures and digit line structures.

35. An electronic system, comprising: an input device; an output device; a processor device operably connected to the input device and the output device; and a memory device operably connected to the processor device and comprising: pillar structures vertically extending through blocks of a stack structure comprising conductive material vertically alternating with insulative material; conductive plug structures within horizontal areas of the blocks of the stack structure and coupled to different groups of the pillar structures than one another, each of the different groups of the pillar structures including multiple of the pillar structures positioned within different sub-blocks of one of the blocks than one another; horizontal sense transistors gated by the conductive plug structures; read source lines vertically overlying the horizontal sense transistors; digit lines structures vertically overlying the read source lines; first vertical read selector transistors vertically interposed between and in electrical communication with the read source lines and source regions of the horizontal sense transistors; second vertical read selector transistors vertically interposed between and in electrical communication with the digit lines structures and drain regions of the horizontal sense transistors; and vertical write selector transistors vertically interposed between and in electrical communication with the digit lines structures and the conductive plug structures.

36. The electronic system of claim 35, wherein the memory device comprises a 3D NAND Flash memory device.